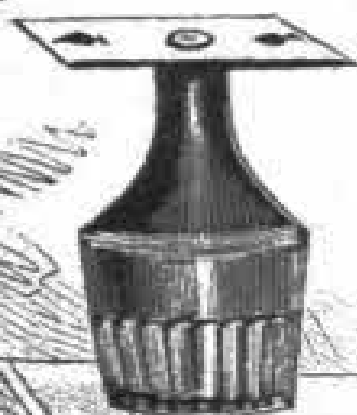
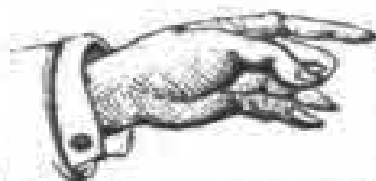
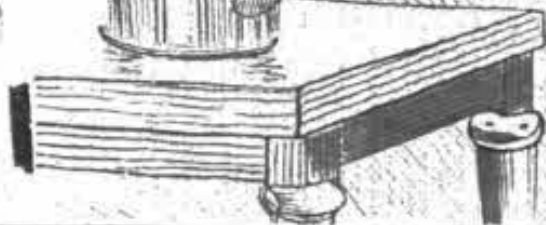


How
To Do
MECHANICAL
TRICKS?



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HOW TO DO

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HOW TO DO MECHANICAL TRICKS.

The Pile of Draughtsmen.

“Matter is inert.” That is what you read in every treatise on physics—what does it mean? Here is a very simple experiment that will prove this truth to anyone.

Pile up ten draughtsmen, as shown in [Fig. 1](#). Before this pile place another piece on edge, and pressing its circumference with the forefinger, let it glide from underneath so that it strikes the pile with considerable force. The piece so thrown must, you will think, upset the whole pile of draughts; but no: the piece thus sharply sent forward will strike only one piece of the pile, and this alone will be dislodged without putting the others out of their equilibrium, and the whole column above will settle down together on the bottom piece.

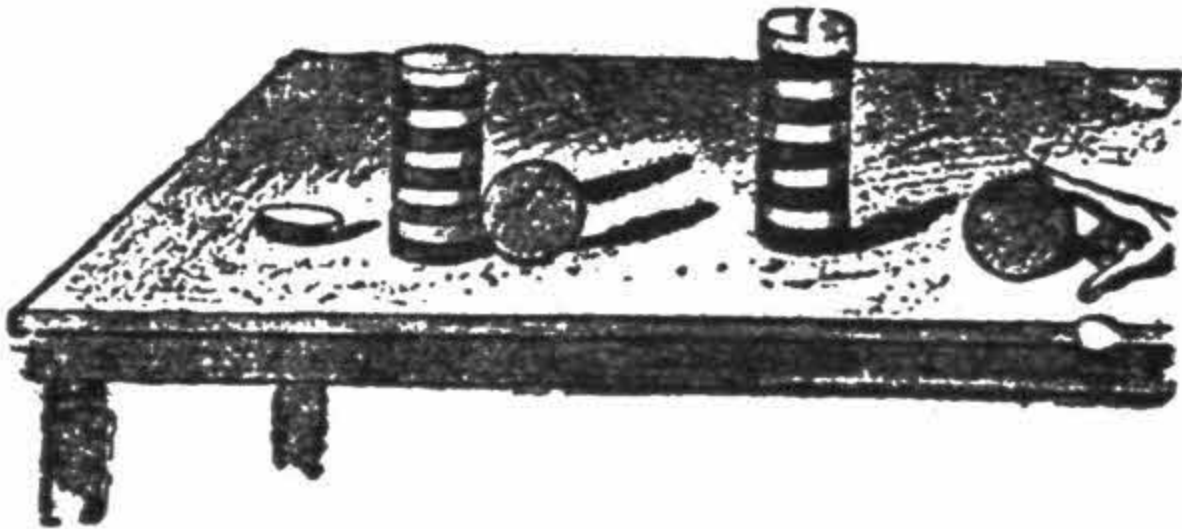


Fig. 1.

In effect, the force of the impulse, making itself felt on the piece that is touched, the latter leaves the pile without transmitting its movement to the other pieces, which, following another physical law, that of gravity, descend vertically to fill the place left vacant.

The experiment may be varied by using a knife and striking with it a sharp horizontal blow on one of the pieces. The piece struck will fall out of the

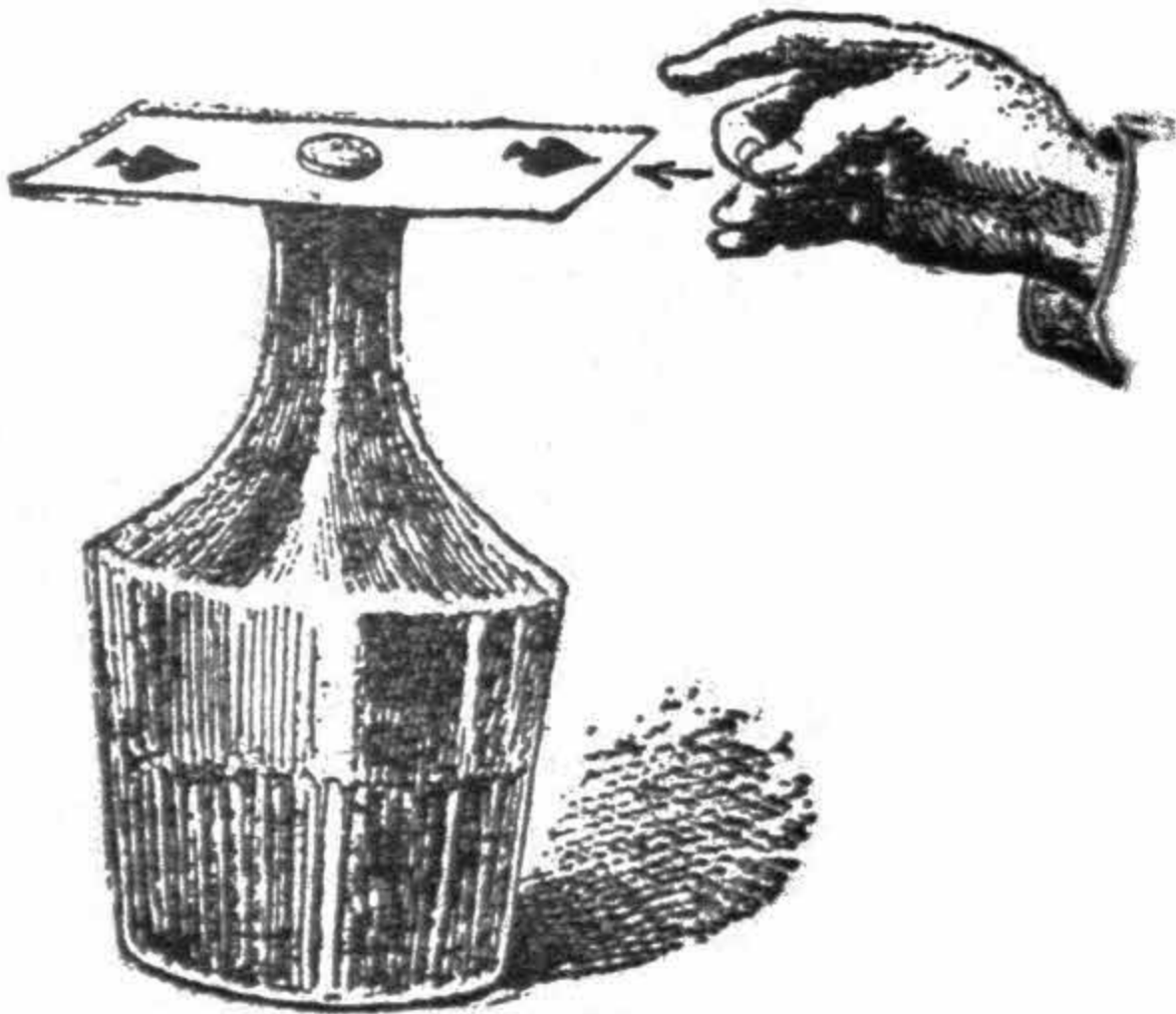
pile without disturbing the symmetry of the others.



The Decanter, Card, and Coin.

This law of “Inertia” will provide us with a few more experiments as curious as they are conclusive.

Place a playing or an ordinary visiting card on a decanter; upon the card and just in the center, over the aperture of the decanter, put a small coin (a dime). Now, if with a sharp fillip, given horizontally on the edge of the card, you succeed in whisking it off (which is very easy), the coin will fall to the bottom of the decanter. The following phenomenon has taken place: the movement was too rapid to be transmitted to the coin, and the card alone was whisked off.



The coin being no longer sustained by the card falls, of course, vertically, without having in the least come out of position.

A sharp horizontal knock given with a penholder or small stick on the edge of the card, will produce the same result, but the fillip is more effective.



A Clever Blow.

Take a thin stick about a yard long, and thrust a pin firmly in each of its extremities. This done, place the stick on the bowls of two pipes, which a couple of persons hold by the stems, in such a manner that the pins only rest on the pipes. A third person then strikes the stick sharply in the middle, and it will break without injuring the pipes.

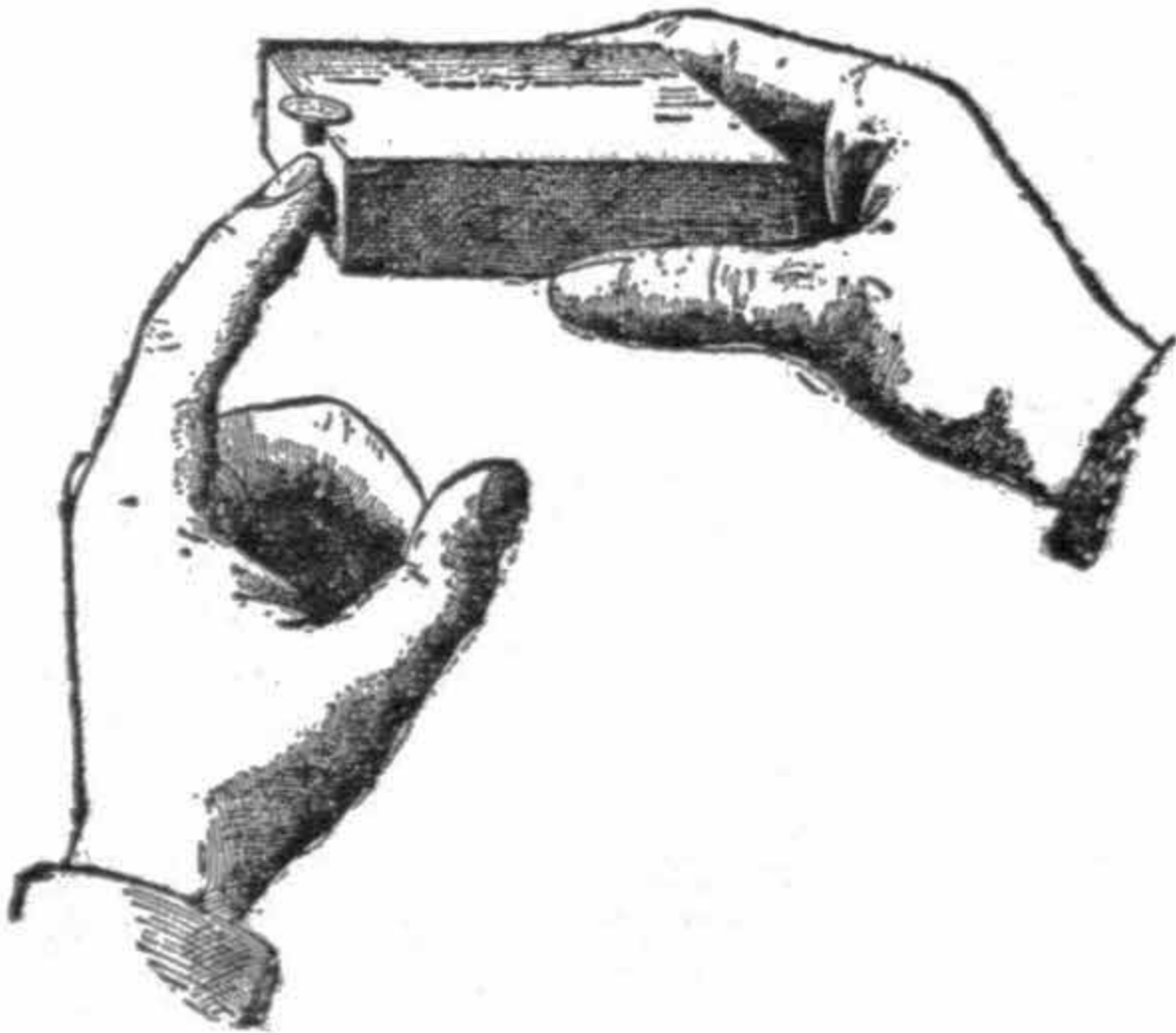
Ordinary clay pipes will do very well, as the more brittle the pipes are, the more striking is the experiment. How is this explained?



The mechanical effect of the shock has not time to reach the bowls of the pipes (inertia), and is only manifested at the very point on which the blow falls, hence the stick unable to resist the force of the blow at the one point breaks in two pieces.

The Obedient Coin.

Take an ordinary wooden matchbox, and remove the drawer holding the matches. In the center place a small coin, a cent will be the best for the experiment, the object of which is to make the coin fall into the interior without touching it. Tap lightly on that side of the box to which you desire the coin to come, until it rests upon the edge.



Then slightly raise the end of the box whereon the coin rests, and lightly tap with the finger once more. At once the coin will fall into the box. The secret of the experiment is this: the taps on the box only move the box, while the

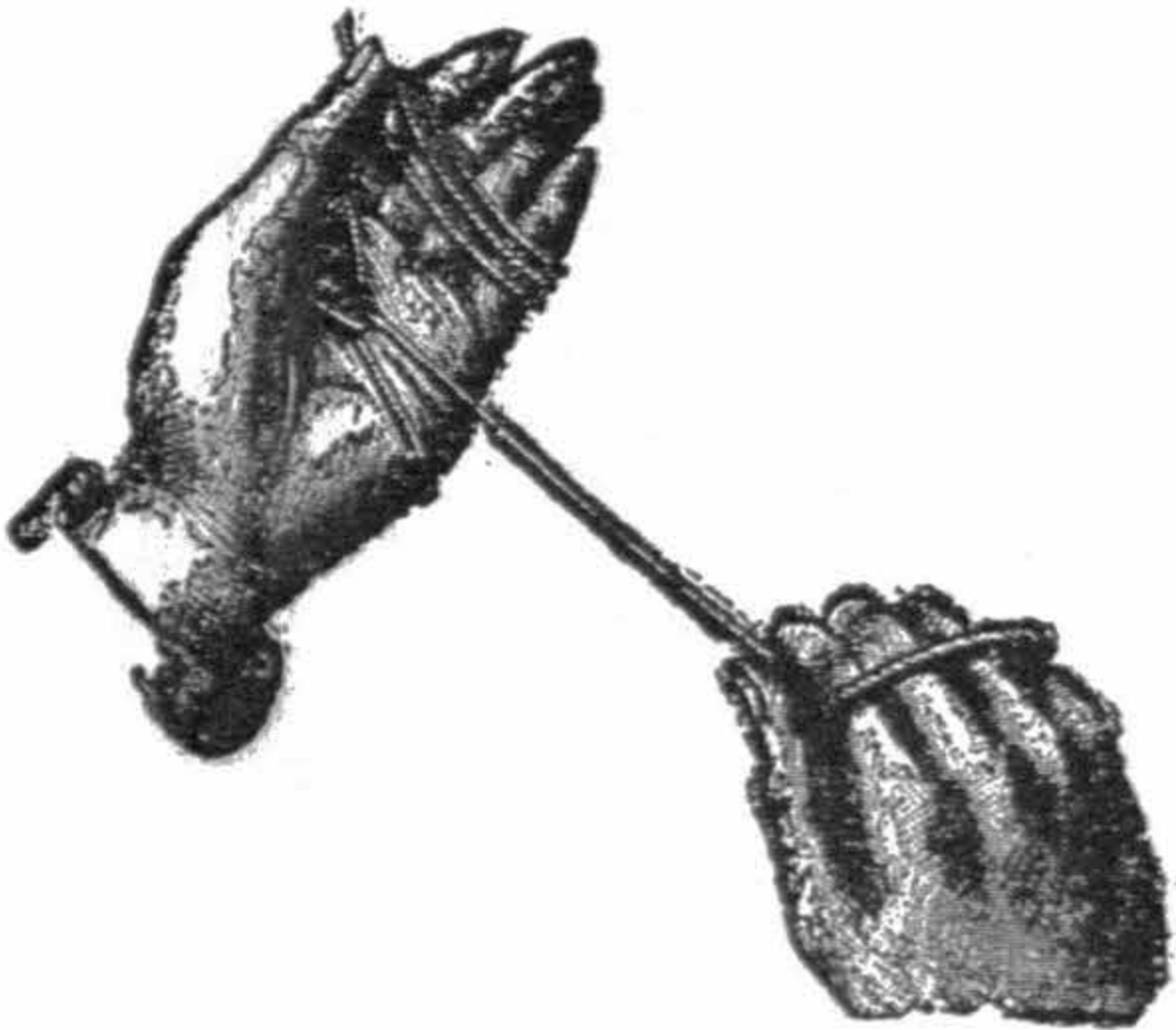
coin retains its position by reason of its own inertia, until the edge of the box reaches it. The last tap knocks away the support, and the coin, obedient to the law of gravity, falls vertically into the interior of the box. This little experiment is easily performed, and extremely interesting when done neatly.



To Cut a String With Your Hands.

With a little practice, and some briskness of movement, you may be able to break a string of considerable thickness by proceeding as follows:

Wind the string round your left hand, so as to make a loop, as shown in the figure. Pass it three or four times round the fingers to insure the solidity of the loop. Seize firmly the other end of the string with your right hand, around which you wind it three or four times, then give a brisk pull. The string will be clean cut at the junction of the loop in the left hand.



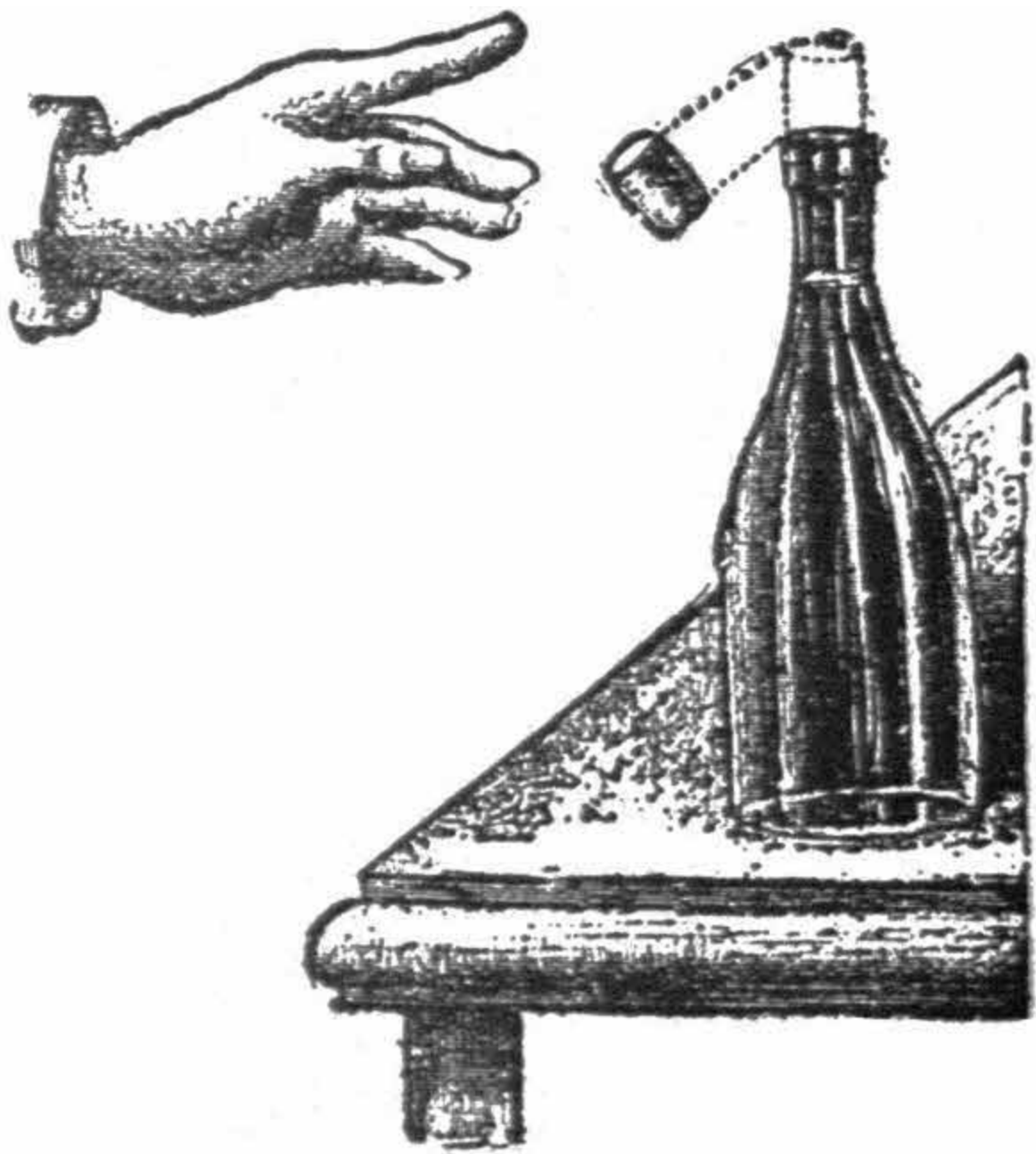
When the knack is well acquired, one may break the string on two fingers only, by following always the same theory as above.

The Rebound.

On the neck of a bottle place a cork in an upright position. The cork must be large enough to rest on the neck without falling in.

Now give a sharp fillip on the neck of the bottle, and you will see the cork fall, not on the other side of the bottle as most people expect, but forward in the direction of the hand giving the blow. This, again, is an illustration of the principle of inertia. A rapid blow tends to push the bottle from the cork before the movement is transmitted to the latter.

Few people will execute this experiment properly the first time, for the instinctive fear to break the bottle and cut their fingers, will prevent them giving a blow sharp enough to make this experiment successfully at the first attempt; but with a little perseverance, the necessary degree of force will be gauged to a nicety.



A Fiery Catapult.

Take a match-box and place it upright edge-wise and place two matches in each side between the inner and outer box, heads up. They must be inserted deeply enough to stick firmly.

Place a third match cross-wise between them and it will stay there by the pressure the latter exercises on them.

Now light the middle of the horizontal match and wait. What do you think will happen? Ask the bystanders which will first catch fire?

The natural conclusion they will draw will be the following.

From the middle the frame will spread of course to the two extremities and light the other two matches, probably this side first where the two phosphorous heads meet.



Well, nothing of the sort happens. When the volume of the burning match has diminished, and consequently its rigidity also, the force of its resistance grows weaker as the combustion proceeds.

A moment comes when the two vertical matches, trying to assume again their original position, throw off, with a sway, the burning horizontal match.

The burning match was rendered flexible in the middle, and is not at all burned at the ends, and the two matches remain standing as before.

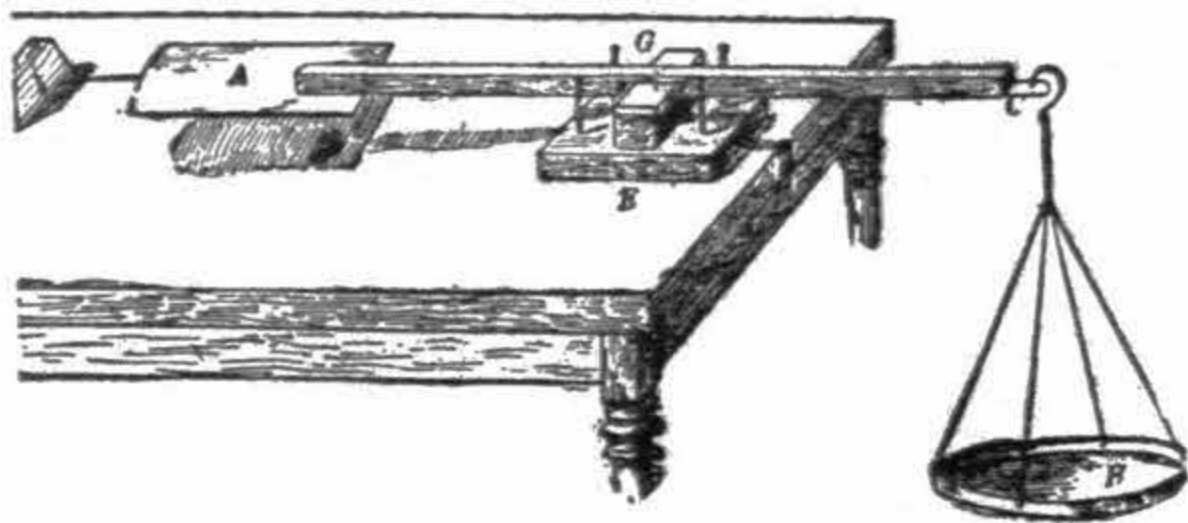


To Make an Exact Balance.

To construct by yourselves, with the help of simple materials a balance of great precision may seem impossible. Nevertheless it can be done.

A ruler, a tin box, (in which blacking was contained, for example) three blocks of wood, two pins, thread, four nails, a small piece of glass, and cardboard are all the necessary materials, and now to work.

At a short distance from the center of the ruler, and on a cross line with one another, stick two pins so that they come out a little on the other side. At one end of the ruler, in C, nail a small piece of your box.



At the spot, where the hook to which the scale is suspended, is to hang, make an indentation with the point of a nail, so that the hook does not shift at the other extremity, in A, fasten a flat piece of tin, which will form one of the scales of your balance.

At the end of this pan solder a pin point downwards. Your second scale, B, destined to contain the object or substances to be weighed, will be formed by the lid of the blacking-tin.

On its rim at nearly equal distances pierce four holes, on which the suspension-strings will be tied, the latter at their upper end being united

together in one string, which is tied to a hook (a bent pin or fishing hook will do.)

Now the point of support remains to be constructed. On a wooden square, rather thick, E, fix another block, G, on which gum a piece of glass. In the largest block knock four nails to prevent the shaft of the balance swerving from right to left.

The small truncated pyramid, D, which you perceive on the left of the design, and which is graduated, serves as bench-mark.

In order to weigh you use the method due to Borda, called the method of double weights.

Place in the scale A a weight which you think is slightly over the one of the substance or object to be weighed. Then the scale B being occupied, get equilibrium by shifting more or less towards the ruler, the weight on the scale A.

Then note the division indicated by the pin point, and take from scale B the article placed there, and put therein weights until the point of scale A tells you that the equilibrium is the same as when the substance was in the scale.

It is not necessary that this balance be exact, provided it answers the very small differences in the pans.

The one we have indicated will weigh down to a fifty-thousandth part of a pound.



The Recomposition of Light.

It is a great pity that exquisitely beautiful facts and mysteries are wrapped up in the crack-jaw terms of foreign languages, and so made to appear ugly.

There is no branch of knowledge more fascinating than light. To follow up its study is like walking along a shady lane, where at certain distances apart the wayfarer lights upon jewels of great brilliance.

It has been said above that white light is formed by the union or combination of seven colors. When a ray of light passes through a prism it is split up into the parts of which it is composed, and seven colors as in the rainbow appear.

These colors shade off into one another with every variety of tint, like a band of rainbow-colored ribbon. This band is called a spectrum.

Now, where science classes are held there may be seen a complicated instrument, which is used to show how the seven colors unite to form white light. It is a disc on which the colors of the spectrum are painted, and it is made to spin round with great rapidity.

The impression received by the eye, when looking at the revolving disc is total abstinence of color. In other words it is white light.

Fortunately, you can satisfy yourselves on this point without any other materials than a cardboard disc and a piece of string. On this disc paint in small sections the colors of the spectrum, repeating them four or five times in the following order: red, orange, yellow, green, blue, indigo, violet.



That the experiment may be entirely successful, the sections must be marked off according to the following scale of width of section. Let orange,

next to the circumference represent 2: then

Red will be represented by	5
Orange “ “ “	2
Yellow “ “ “	5
Green “ “ “	4
Blue “ “ “	5
Indigo “ “ “	3
Violet “ “ “	5

Now, in any diameter of the disc bore two holes not too near the edge. Through them pass a piece of string, and knot the two ends together. Take hold of the string with both hands, and make the disc spin round.

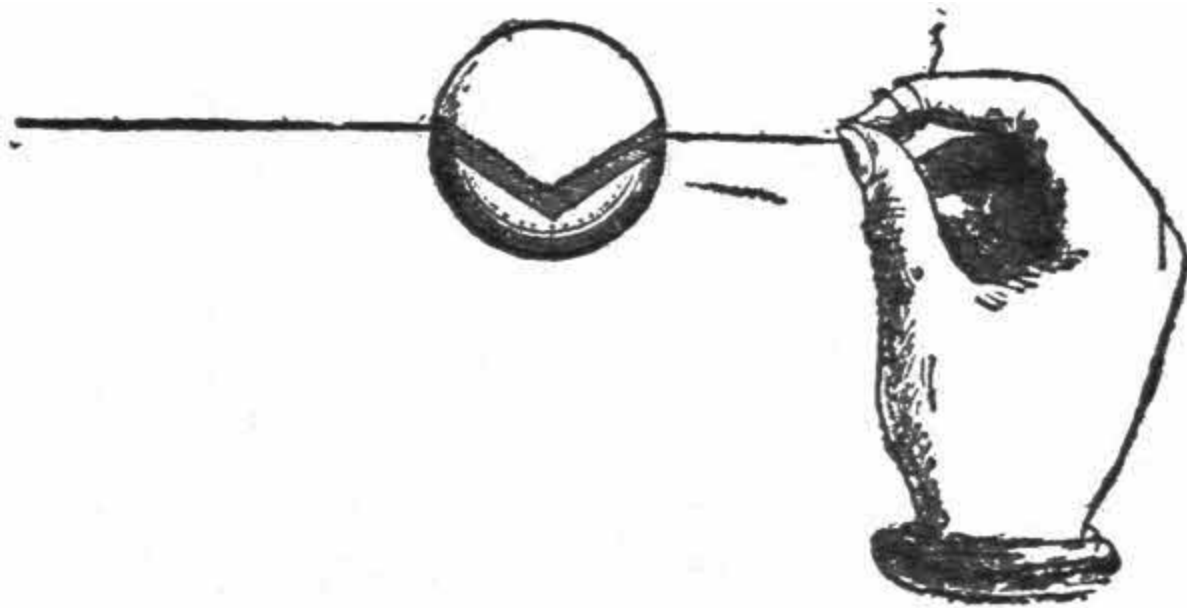
Then extend and approach the hands alternately to give a very rapid movement to the disc. When revolving rapidly enough you will not be able to distinguish the separate colors. They all become blended into white light.



The Mysterious Apple.

Pierce an apple in such a manner as to obtain two holes tending toward the middle, and forming a pretty large angle as shown in the figure. Two quills or tin tubes should be inserted to make the inside passages smooth. Pass a string through the hole and your apple is prepared for a little trick, which, you may be sure will astonish all persons before whom you practice it, and who of course are not yet initiated.

You fasten one extremity of the string to your foot, and take the other in your hand so as to produce at will the rigidity of the string. You can then command the apple to go down, or to stop, and it will obey your order immediately. Indeed, when you straighten the string, the part which enters the apple pushes against the angle formed by the two passages, and by the pressure, holds the apple. When on the contrary you let go a little, you take away the rigidity and the apple glides down.



You can therefore alternately let the apple go down or stop its course, and we repeat it, persons not in the secret cannot imagine by what means you get this curious result.

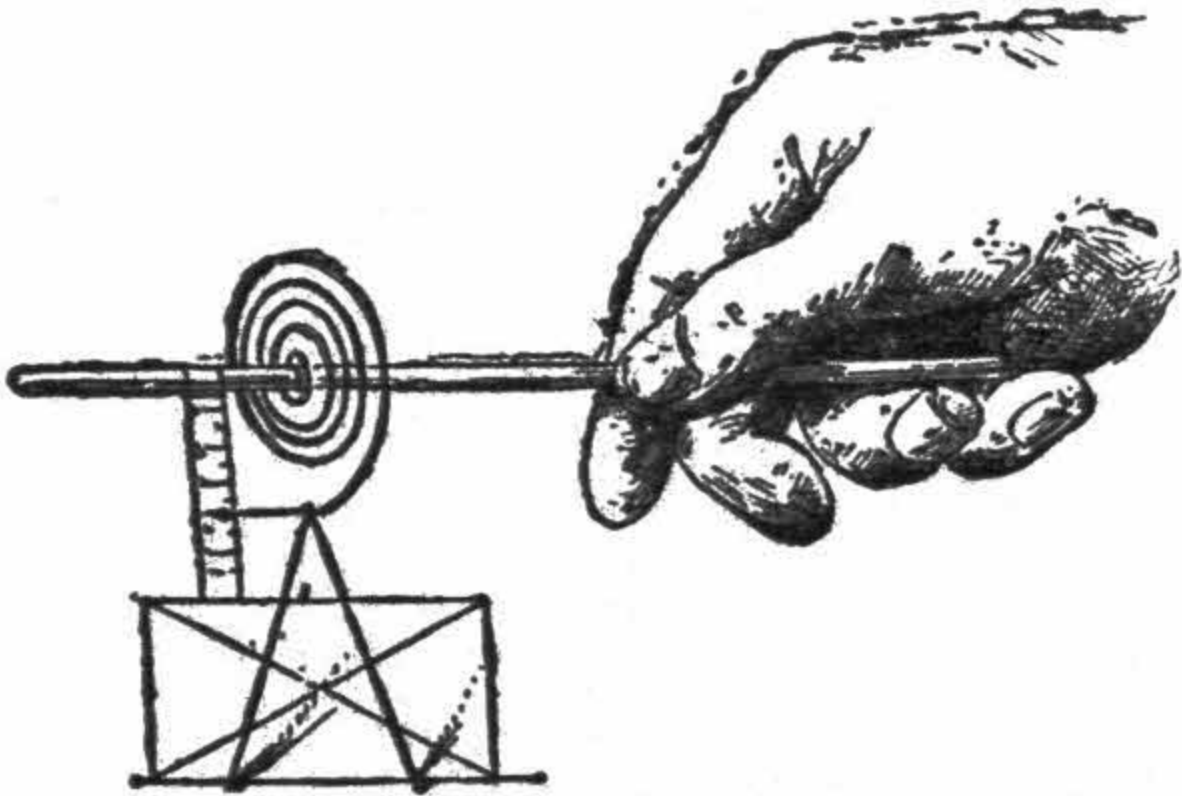
If, instead of an apple one takes a wooden ball, the experiment will be more interesting and the article will last longer.

Economical Letter-Scales.

Take a watch or small clock spring, and fix it by the center on a stick. At the other end attach a small brass hook to hold letters, etc., as shown in the figure.

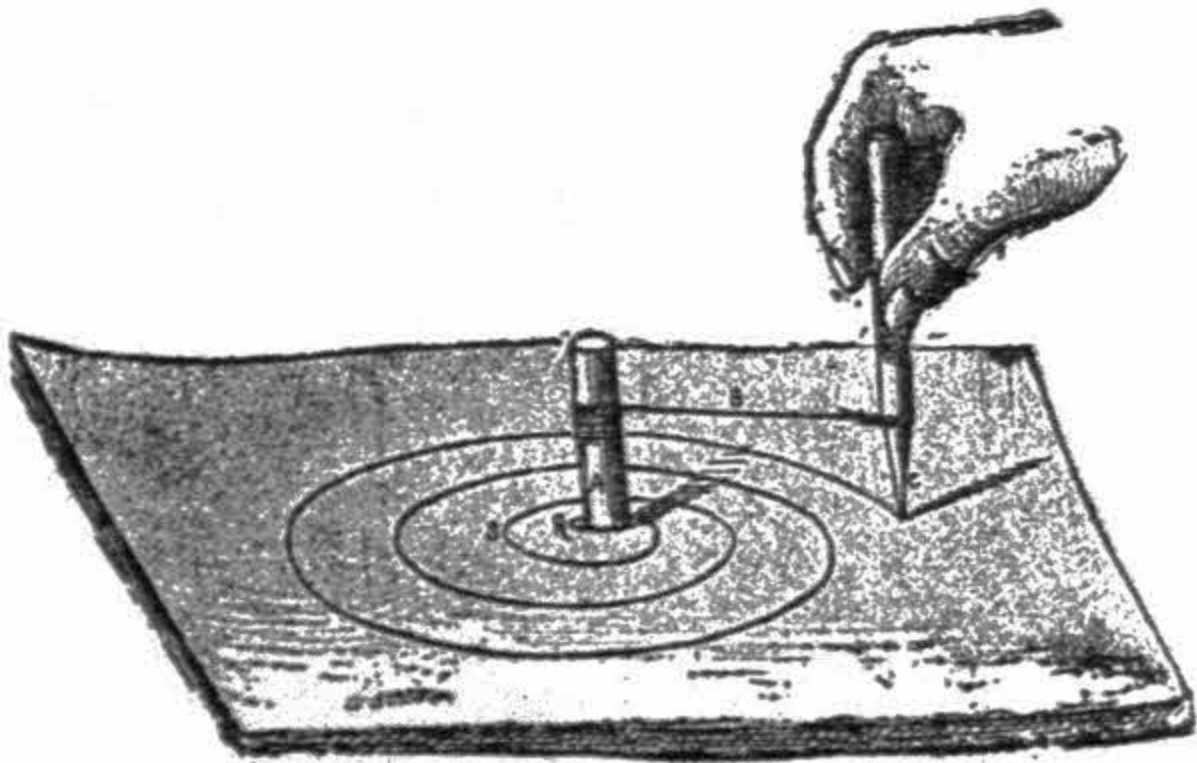
At the top of the hook fix horizontally a small band, running over a strip of cardboard, likewise hanging on the stick.

Now graduate the cardboard strip with real weights, or their exact equivalents, and after this any small articles may be weighed with sufficient accuracy. The spring, being of steel, always turns to its original position when the scale is empty.



Tracing a Spiral.

In geometry the process for tracing a spiral by the help of compasses is pretty long and tedious. The following method will enable you to do it far more quickly and as accurately.



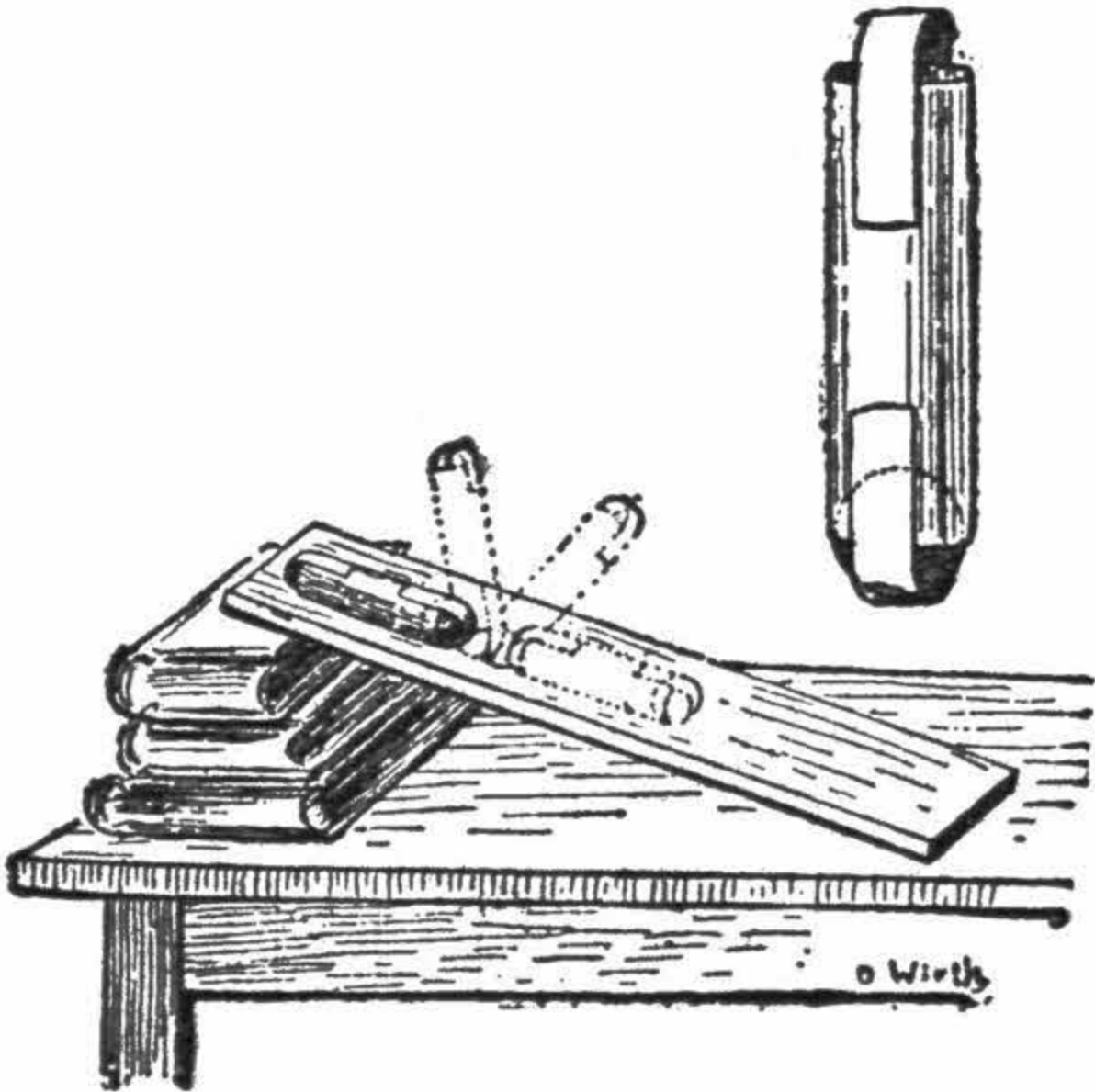
Take a wooden or cardboard cylinder, with a diameter equal to a fourth part of the distance you require between the spires (or trelices) to be traced. On this cylinder fasten one end of a string, B, and wind it up, and attach to the other end a pencil, C, or a point, according to what you want to do.

Now you have only to turn to right or left according to the direction in which the string was wound up, by holding the pencil down and keeping the string tight, and a spiral of perfect regularity will be traced.

The above figure clearly shows the process. The cylinder A has a diameter equal to the distance R S divided by 4.

The Inclined Plane.

Take a piece of paper, roll it up into a tube large enough to hold a marble, and gum it lengthwise. Then introduce a marble and close the extremities with a strip of paper as shown below.



When you think that it is well dried you place it upright on the upper end of an inclined board, or flat ruler, leaning on a pile of books for example. You

will then see the paper cylinder lie down, get up and so on till it reaches the bottom of its course.

The effect is very curious and will be more so if you are somewhat of an artist, and able to draw or paint a figure on the cylinder.



To Cut a Bottle With a String.

Gum first two circular pads of paper on each side of the spot where you intend to cut your bottle. These pads are obtained by gumming several strips of paper one over the other, so as to leave between them a groove on which you wind the string round once.

Catch hold of the extremities of the string, and draw it to and fro, see-saw fashion, by which friction the part of the glass operated on will be heated.



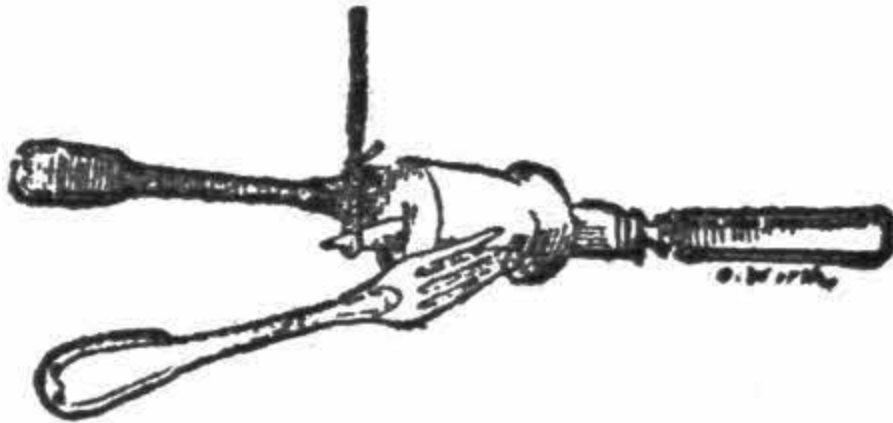
As soon as you think that the glass is hot enough, plunge the bottle in cold water, which you will have placed handy before, and at the spot where the friction was exercised the glass will be clean cut. According to the thickness of the glass, more or less heat must be produced. This process is infallible.

The same result can also be produced in another way. It is, when once the heat is sufficient to let glide a few drops of water along the string. The string must be well wetted. The cut will be as clean as by the other process.



Equilibrium of a Knife in Mid-Air.

Be reassured dear readers, we are not going to ask you to make a balance in mid air, that would be too much for our weak capabilities. The question is simply to swing a knife horizontally in the space which surrounds us. The experiment is curious and easily executed.



Take the cork of a champagne bottle. Pierce it lengthwise with a sharp knife, and let the knife stick out a third of its length from the thin end of the cork. Then insert into each side of the cork the prongs of two forks, so that they are perpendicular with the blade of the knife as shown in figure.

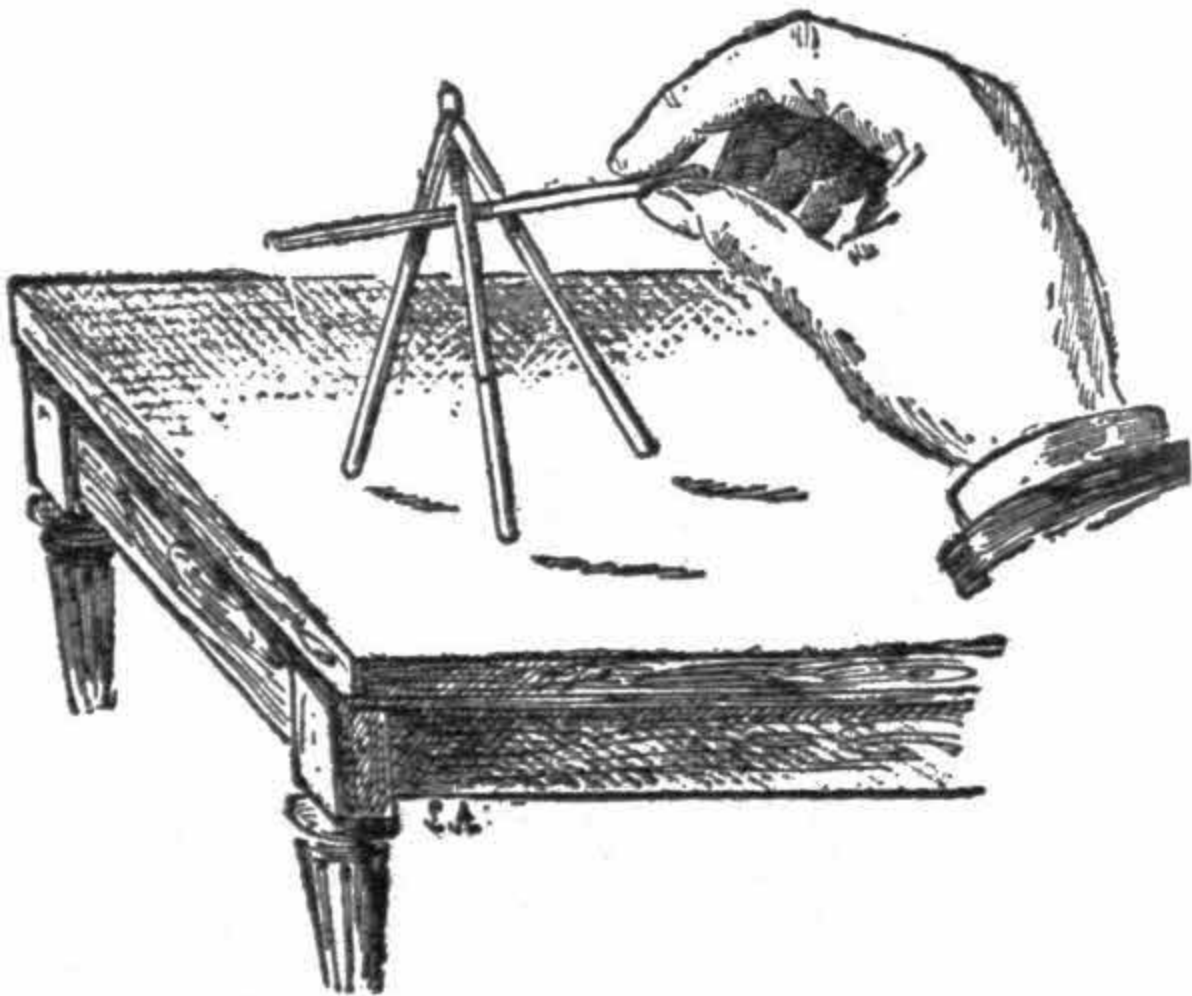
This operation accomplished, you have only to suspend the point of the blade on the loop of a string, and the knife will hang horizontally. You may then swing it if you choose, and the movement will not destroy the equilibrium.

A Trick With Four Matches.

Speaking of matches, there is yet one more trick to be played with four of them.

At the non-phosphoric ends of two matches cut a small notch so that they fit into each other. Stretch the matches apart so as to form an angle, and place them vertically upon the table. Then lean a third match against them so as to form a tripod, standing by itself.

The question now is to take up this trivet with a fourth match and carry it to another place without disturbing the harmony of the little construction.



At first sight this seems impossible; it is, however, easily done. You have only to slide the fourth match between the two stuck together, and the one serving as support.

By lightly pressing against the two first ones the third one will slide, and its upper extremity will come between the angle formed by the two others. By taking it up briskly, this extremity will be maintained, and you are then enabled to carry the little tripod to another place.



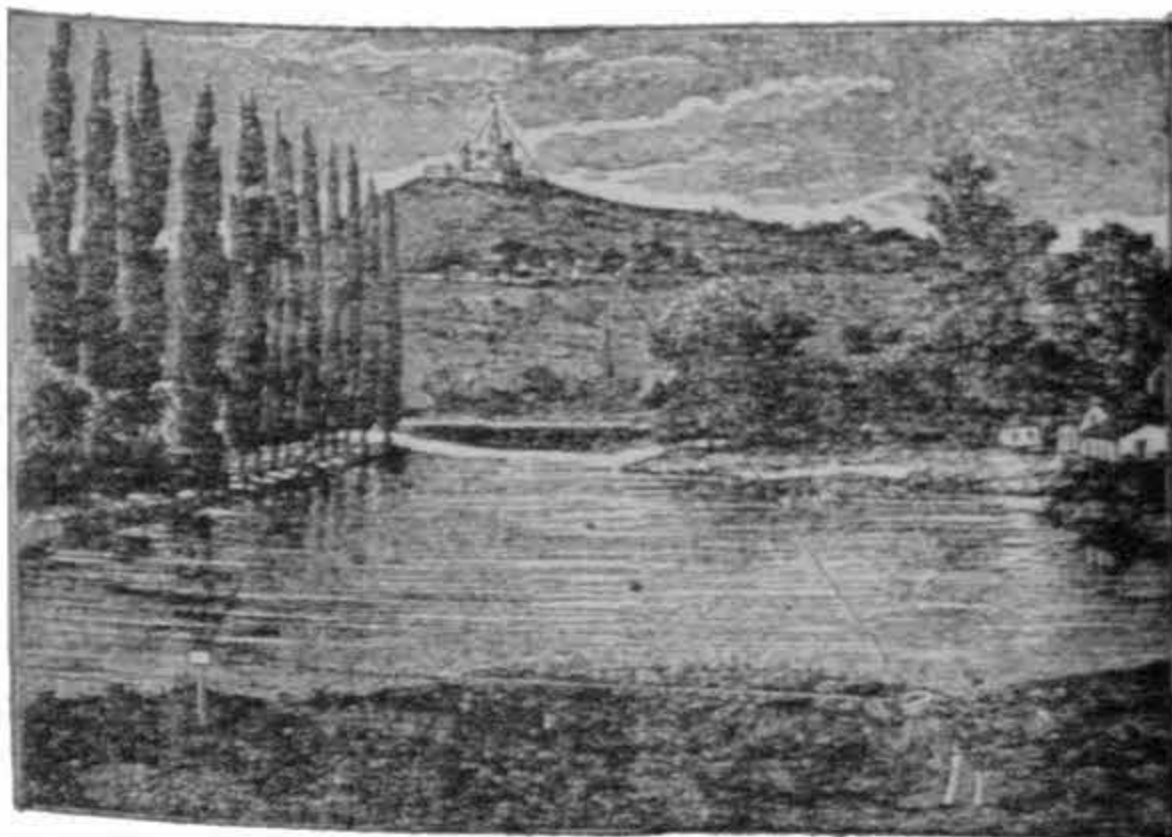
The Distance of an Inaccessible Point.

Everyone knows what an angle is, and you say at once it is the inclination of two lines that meet each other. These lines by their branching off form an opening more or less wide. This opening is measured by the aid of an instrument called a *protractor* made of brass or horn, which finds its place in nearly every box of mathematical instruments.

It represents a semi-circumference, divided into 180 equal parts, called degrees, written thus: 180° . Each degree is divided into 60 minutes, expressed thus: 60 min.; and finally the minutes are divided again in 60 parts, called seconds, indicated thus: 60 sec. There are therefore in a whole circumference, 360 deg., 2,160 min., and 12,960 sec.

One degree, therefore, is the 360th part of a circumference, and thus we have a measure independent of all dimensions. For example, on a round table of 36 yards in circumference, one degree will be marked by one tenth of a yard; on a pond of 360 yards in circumference, one degree will be equal to one yard.

The degree, therefore, may be more or less, but it is always the 360th part of the circumference of a circle. Let it be quite understood that, whether an angle is to be on a sheet of paper, or in the skies, the divisions do not change.



This must be well grasped, it is of the utmost importance for the explanations which follow. It is therefore settled: the measure of the angles has nothing to do whatever with a measure of length.

We have shown how to measure an angle. Let us examine now what is a triangle, without pondering too much over this geometrical figure, which every one knows. The essential property of this three-cornered figure is that the sum of its three angles is always equal to 180 degrees.

In other words, the protractor placed successively at each angle will give three numbers, which, added, make up 180 degrees. Keep this property well in mind, as it will serve us hereafter.

Now, to what distance does a degree correspond? For example, take a yardstick, and with the *graphometer* (an instrument by which angles are measured), in readiness, carry it from the latter instrument to a certain distance, till the two extremities of the yardstick measure one degree; this yard is then said to subtend an angle of one degree.

Now, measure the distance which divides the yardstick from the instrument, and you will find it to be 57 yards. Therefore, one degree corresponds to an object being at a distance of 57 times its height. A man two yards high at a distance of 57 times his height, or 114 yards will measure one degree.

One minute will be represented by a piece of cardboard of a hundredth part of a yard long seen from a distance of 34 yards; and finally, a second will be given by a card a hundredth part of a yard seen from a distance of 2062 yards.

A hair seen at 20 yards about represents a second. This perhaps, you think to be too small to be seen by the naked eye.

Suppose that you to measure the distance of a church situated on a height, and from which you are separated by a river (see fig.) Choose on the river's bank two spots from which the steeple C can be seen, say A and B. At B plant a surveying-staff, and with the graphometer, go to A and find the angle formed by B A C.

Suppose for example, it reads 84 degrees. Repeating the operation at B for the measure of the angle C B A, suppose it to be 95 degrees. Measure the distance from A to B and let it be 80 yards.

Now here is the statement of our problem:

How to resolve a triangle of which the base is known to be 10 yards, and two of its angles. Well, we have said above that the sum of the three angles is always the same, equal to 180 degrees, having on one side 84, and on the other 95, that makes together 84 by 95, equal to 179 degrees. The difference between this number and 180 is 1 degree, therefore the angle ABC measures one degree.

We know that an angle of one degree corresponds to a distance of 57 yards. Multiply the base of our triangle by 57 yards and you obtain a distance of the church from the points A and B, 10 by 57, equal to 570 yards. Nothing is more simple than this.



The smaller the measured angle the further off the object will be. As seen in our figure, the upright lines, $m o$, $m' o'$, $m, o,,$ do not vary, but according to their distances from point C, they form various angles, ac , $a'c'$, $a,c,,$ becoming smaller and smaller.

A graphometer is not always to be had. When approximate distances only are required, the following contrivance may be used. Trace on a cardboard of large size a semi-circumference which one divides first into 180 equal parts, then each of these is divided again in 2, 3, 4 divisions, etc., according to the size given to the circumference, which constitutes a large protractor.

To measure an angle place the cardboard upright in an horizontal position, supporting it by the center of the semi-circumference by means of a screw fixed on a stick. Then proceed as stated above.

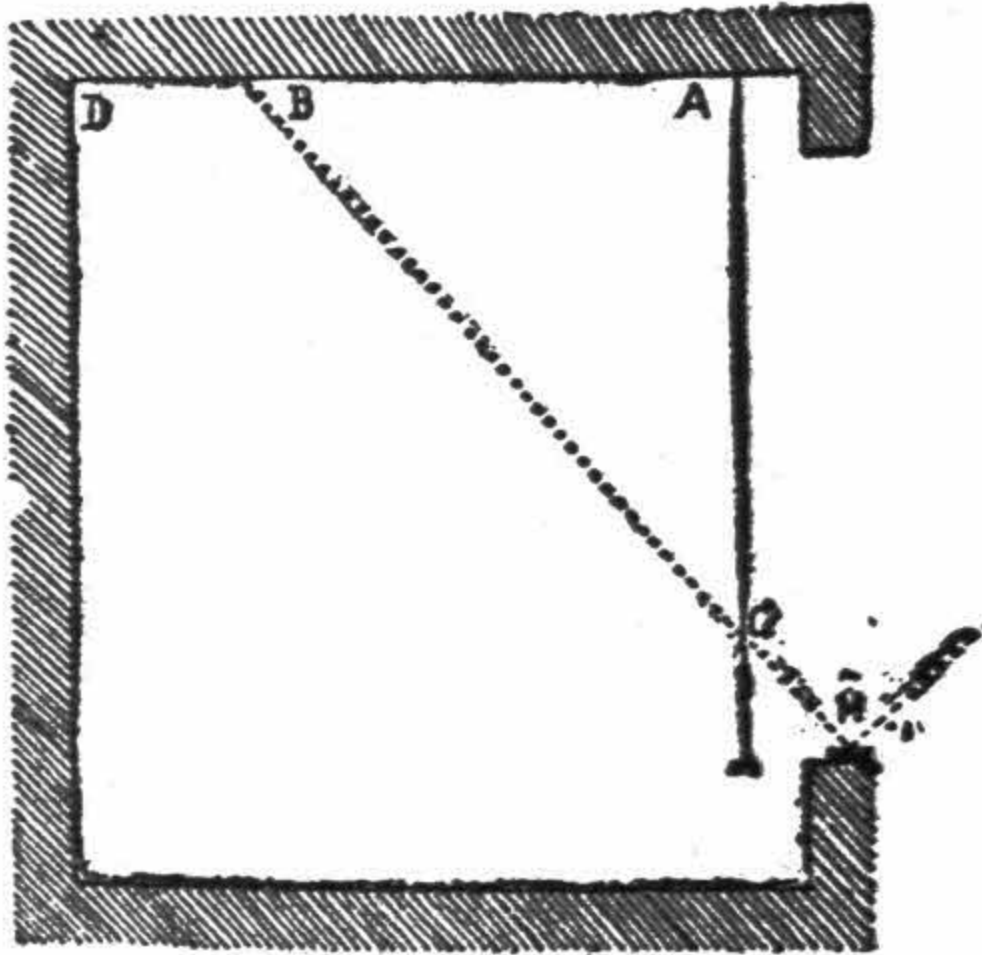
From a pin stuck in the center mark the spot where the visual ray passes, go to A and to B, and you get approximately the desired result.

Practical Tracing of a Meridian Line.

The meridian line of a place is an imaginary line passing through this place and the center of the sun, when the latter is at the highest point of the arc of the circle, which it daily describes. At that very moment it is noonday exactly at the place in question.

As the position of the earth changes from day to day, the sun does not every day touch the meridian line at noon; sometimes it is in advance, sometimes behind.

Various instruments have been invented to indicate in a practical manner the meridian of a place. We owe the following construction to Mr. E. Brunner of the longitudinal office.



On a window-sill in a southerly position, fix in a solid, permanent manner, a small cupful of quicksilver; cover it with a lid made of varnished metal, and pierced in its center by a small round hole about a quarter of an inch in diameter. This lid must fit well, but not too tightly, so as to permit its being lowered in close proximity to the surface of the quicksilver.

When the window is open the solitary ray reflected on the mercury will be projected on the ceiling of the room. At the exact noonday the center of the mirror and the center of the reflected image are in the meridian plane. It remains only to be traced.

At the moment of its passage one marks in B, for example, a spot corresponding to the center of the reflected image; one knocks a small nail there, and with a string connect this point with another outside the window, so that the string passes through the center of the diaphragm, M. The line, B

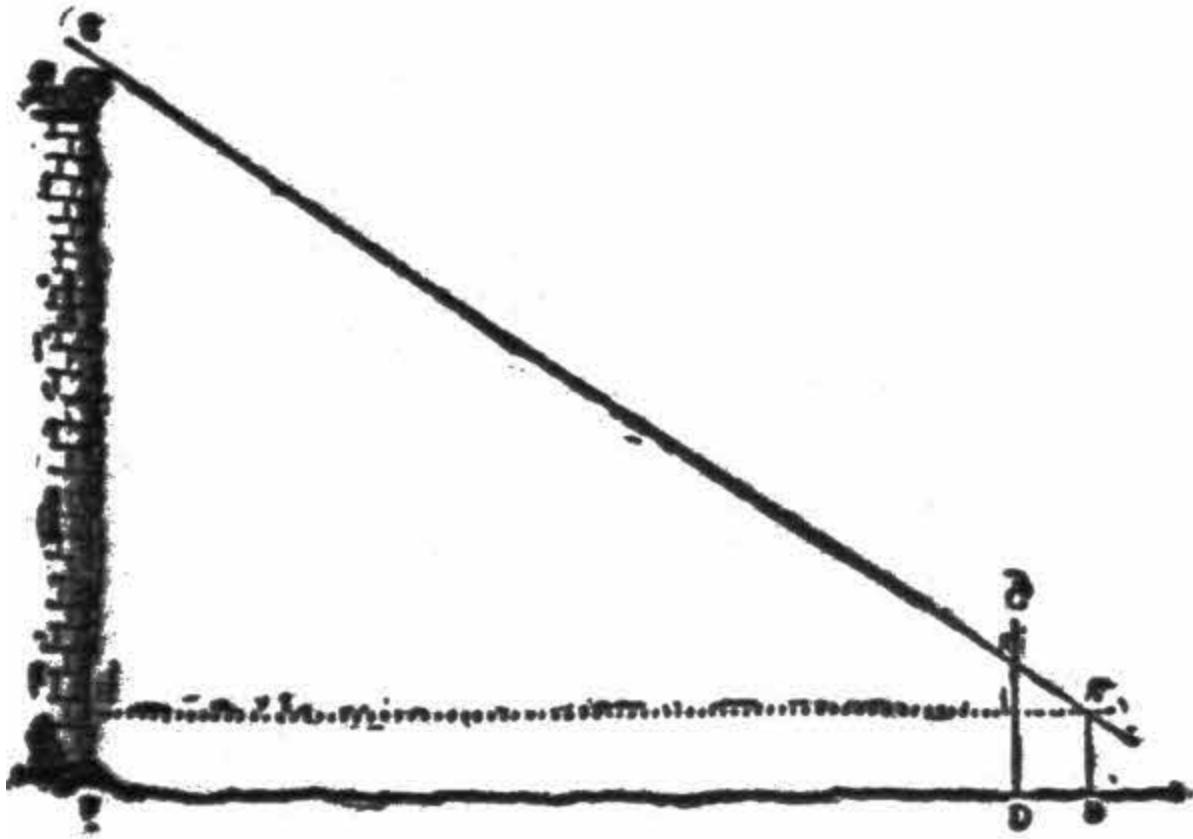
M, is the meridian plane. From A, suspend a lead-line which meets the string, B M.

All you have to do now is to join on the ceiling the points, A B, and continue them to D. A black thread may be stretched to serve as the line, and this is the meridian required.

To get the mean time one has only to note the exact passage, and deduct the corrections given in various astronomical papers.

To Measure the Height of a Mountain.

One can, without instruments, take the height of a building or a mountain, provided you are able to measure their base. A yardstick and two ordinary sticks are enough. Suppose the height of the tower, E F, is to be taken.



Some distance off plant a stick, a yard high, A B; one yard from this we plant another and longer one, C D. Measure exactly the distance, B F, and applying the eye at A, we aim at the summit of the tower, E; mark on the stick, C D, the point where the visual ray meets the stick, *i.e.*, point G.

Then, by measuring the distance, D G, and subtracting one yard you get G I, and may be expressed in the following statement:

$$A H : A I :: E H : G I$$

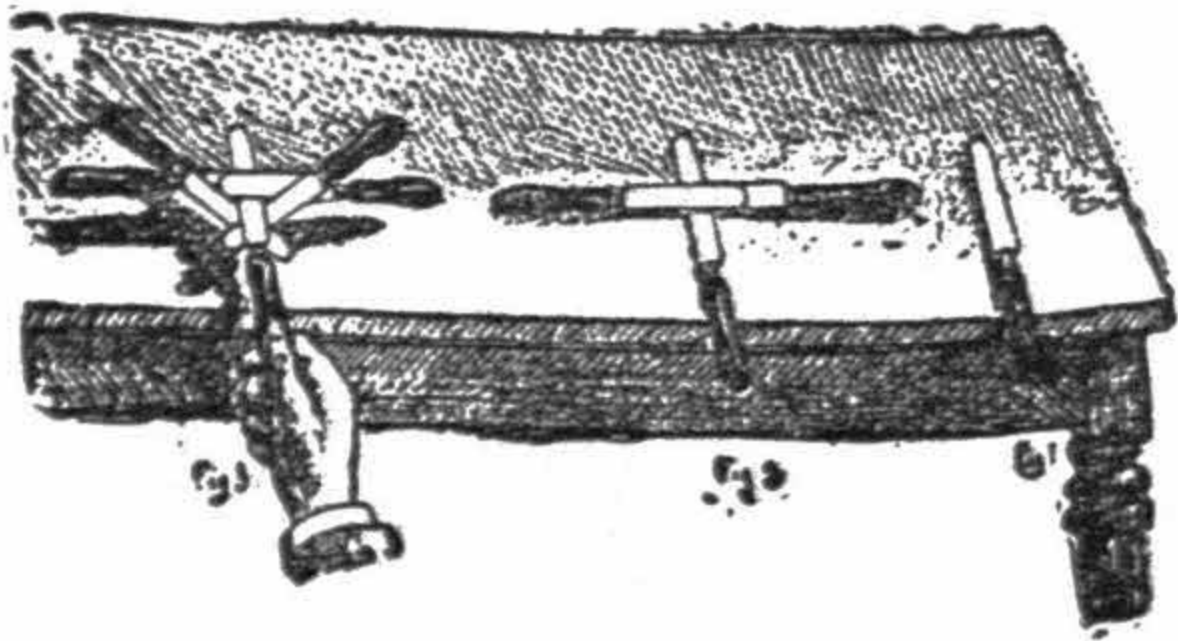
In the given example let us suppose that $AH = 150$ yards, AI will, of course, be equal to one yard; $GI =$, say four fifths of a yard; the problem will be: $150 \text{ yards} : 1 \text{ yard} :: x : \text{four fifths of a yard}$. Work the sum out, and the value of x is 120 yards.

Having taken our lease, AH , at one yard from the ground, we must add one yard to 120, making 121 yards, which is the height of the tower wanted.

To Take Up Four Knives with One.

Here is one more trick of equilibrium, which appears to be interesting enough to find a place among these experiments.

We need not give any long explanations, for our figure fully illustrates the way in which it has to be executed.



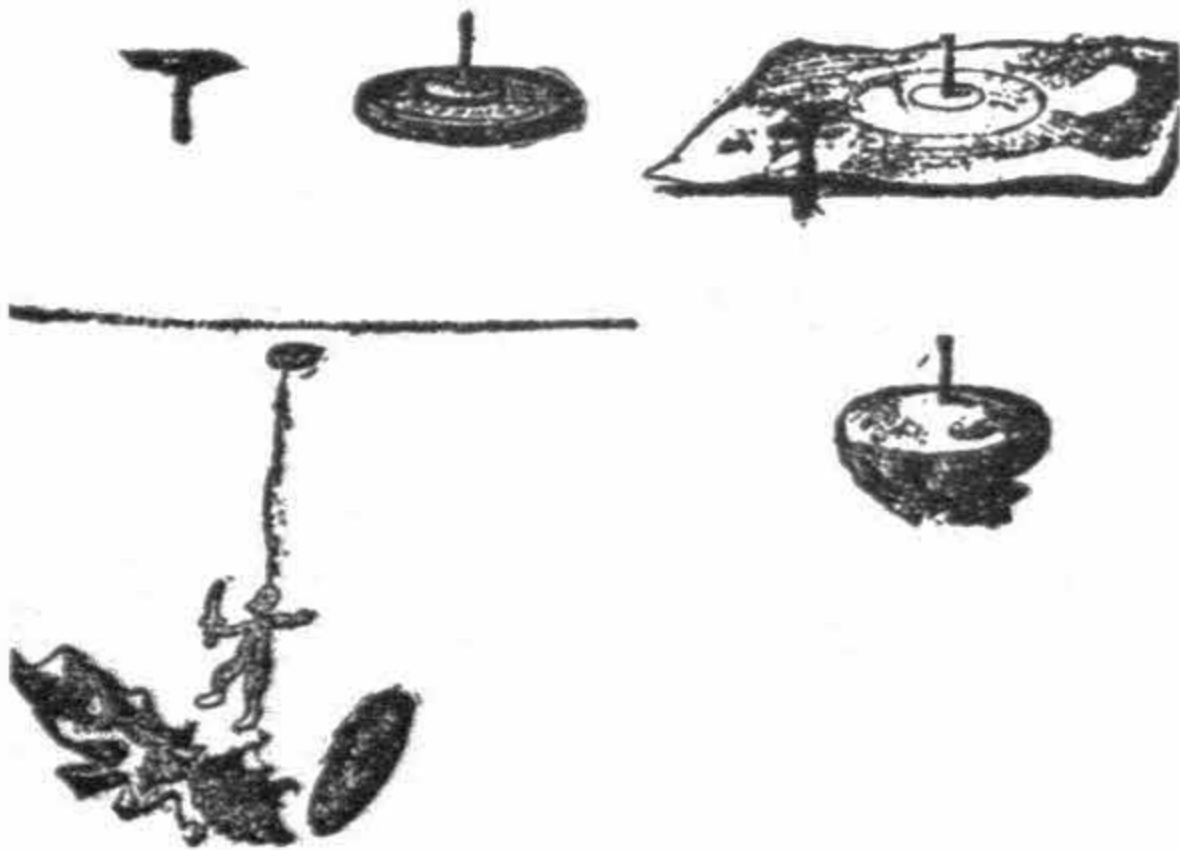
First place a knife straight before you, then two others which you place, blade upon blade, over the first. Finally, the two last ones are arranged transversely, their blades passing over those of the two knives put down in the second instance, and below the blade of the first knife.

By taking hold of the handle of the first knife, you can lift them up all at once without breaking the equilibrium.

The Tack in the Ceiling.

To nail a tack in the ceiling without hammer, using a ladder or chair to reach it, seems as impossible as pulling the moon down from the sky. Yet, with a little cleverness, it is quite an easy thing to do.

Place a tack, head downwards, on a half dollar, then place a small piece of tissue paper over it, so that the point of the tack passes through.



Then turn the sides of the paper down round the coin. Throw the whole, point upwards, violently against the ceiling, trying to keep this projectile of a new description from turning over on its course.

With a little practice the knack is soon acquired. The tack enters the ceiling, the violence of the shock tears the paper, which, carried away by the coin, falls to the ground.

Suppose you have a light object to suspend on the ceiling; you may do it in this manner without much trouble. Simply tie a thread to the tack, the object being attached to the other end.

If the projectile is well thrown the tack will go right in, and stick very firmly.



The Jumping Pea.

Take an unbroken straw, four or five inches long, not closed by knots, but forming a tube, and about one twentieth of an inch in diameter.

Divide one of its extremities to a length of about half an inch in four, five or six parts, which separate slightly, so as to form a truncated cone.

After having thus prepared the straw, take a dry pea, with a larger diameter than that of the tube, and place it in the cone. Hold the tube upwards, and blow into it at the opposite end.



The pea will be forced upward by the air column which you blow into the tube. It will remain suspended in the air as long as the interior pressure continues, then fall back into the arms of the cone.

To vary that experiment pass a long pin through the pea, the point of which is turned into the tube. When well thrown up, the pea can be maintained at a distance of two or three inches from the mouth of the straw. According to the stronger or weaker blast of breath, the pea will go up or down.



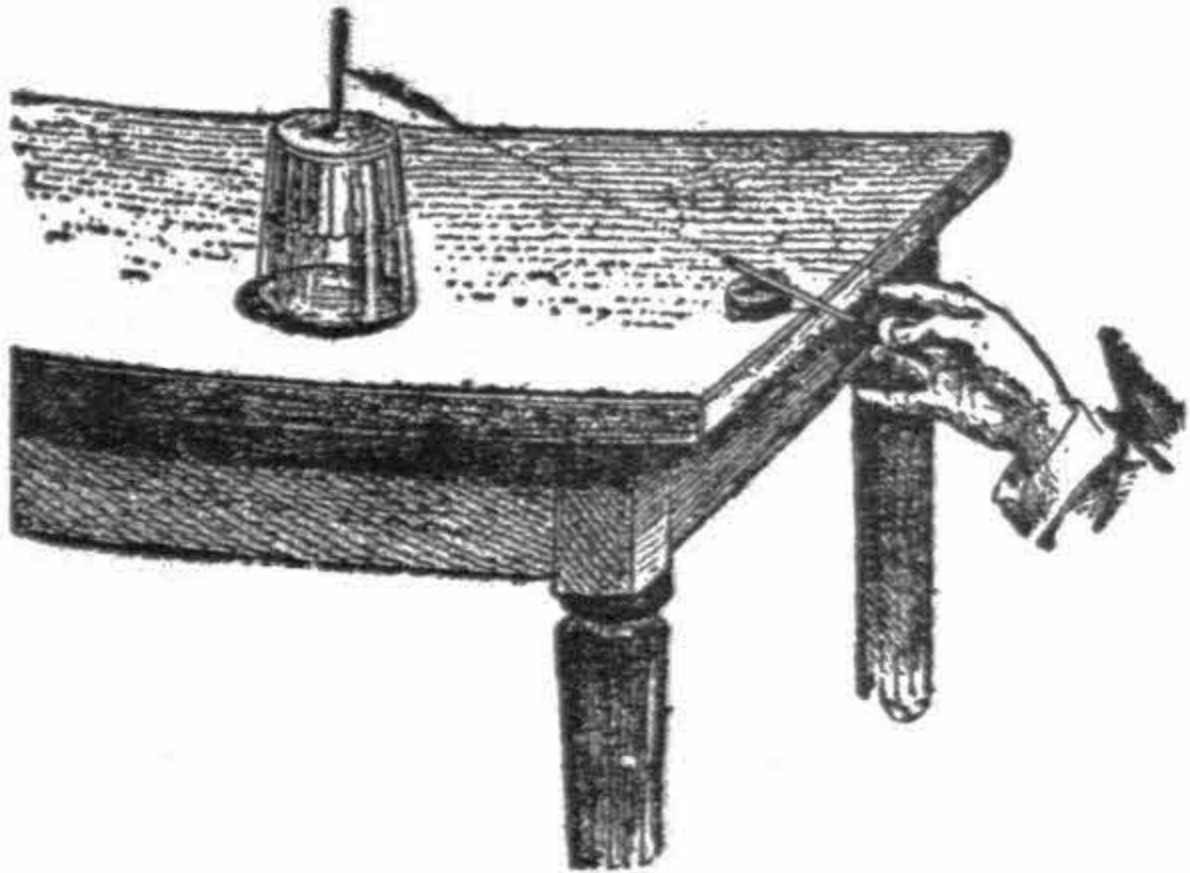
To Acquire a True Eye.

Here is a peculiar and clever recreation, easily performed, though at first sight it may appear difficult.

Put a tumbler upside down. By means of bread crumbs, fix a match vertically on the top. On the edge of the table place another match, partly raised on a piece of cork or wood.

Stoop down and aim at the vertical match on the glass, so that the one on the table is in the exact line of fire.

When you think it is aimed straight, give it a fillip on the lower end, it will shoot up and touch the one placed on the glass if the aim be good.



If you succeed, you may congratulate yourself on having good eyes—a very desirable gift if you should have to handle a gun, as a soldier or a

sportsman.



The Air-Tight Stopper.

How many times has it happened to you, when wanting to cork a bottle, that the intended cork was too large to enter the neck?

What have you done? Cut the cork all round, and obtained, but imperfectly, the desired end.



Next time when the same occasion arises, turn the difficulty in this way: Instead of attacking the sides, cut four notches, bevel-shaped, into the cork as shown in the figure.

Treated in this manner your cork will fit, and close the bottle hermetically.

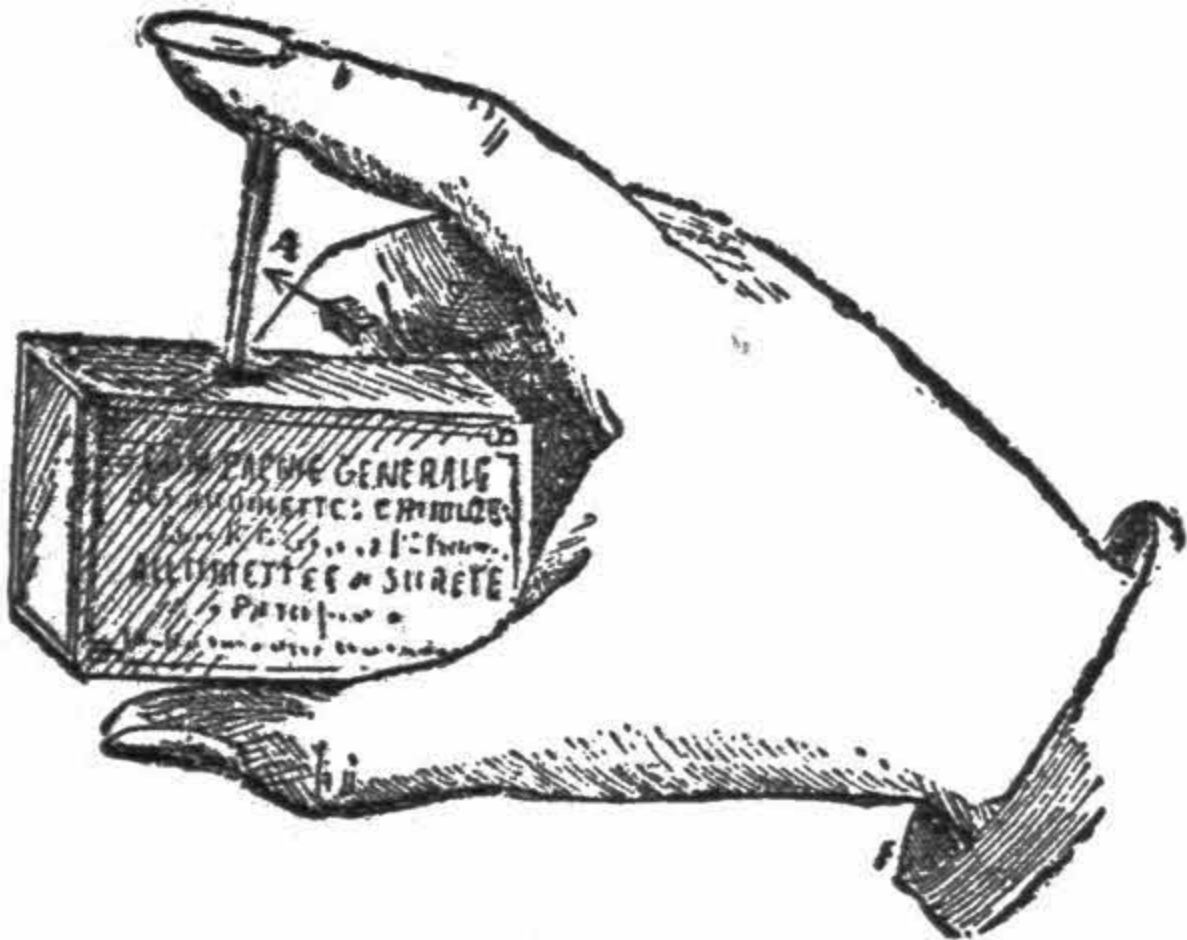
The Fusee Rocket.

For this you only want a simple match box. Take out a match, and holding it on to the box as shown in figure, *i.e.*, hold the box a little slanting, between the thumb and forefinger, and place the match head downwards on the side of the emery paper, where the match ignites when rubbed against.

With medium force press on the match and with the other hand give it a fillip in the direction indicated by the arrow.

The little missile will fly into the air all ablaze, and fall down at a distance of four, five, or even six yards.

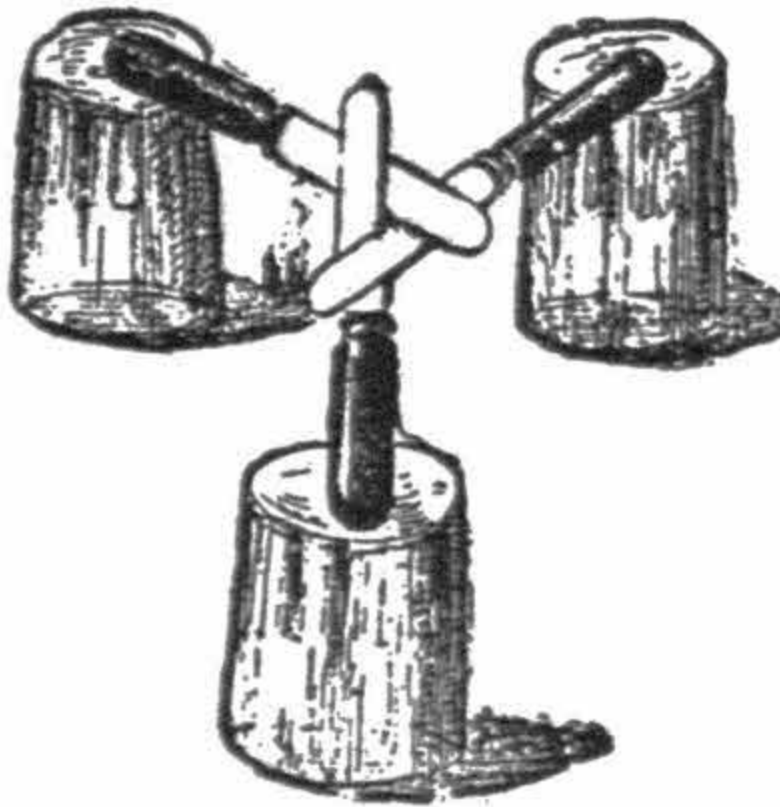
With a little practice you will succeed each time. It looks like a small rocket, especially when done in complete darkness.



Be careful to make the experiment only where there is no danger of setting anything on fire.

A Novel Table Mat.

To construct this original table mat 6 objects, always at hand when table is laid for a meal, are required; 3 knives and three tumblers of equal size and arrange the tumblers upside down, in the form of a triangle, and on each of them rest the handle of a knife. Cross the blades so that the first laid passes over the second and the second over the third, this latter passing over the first X.

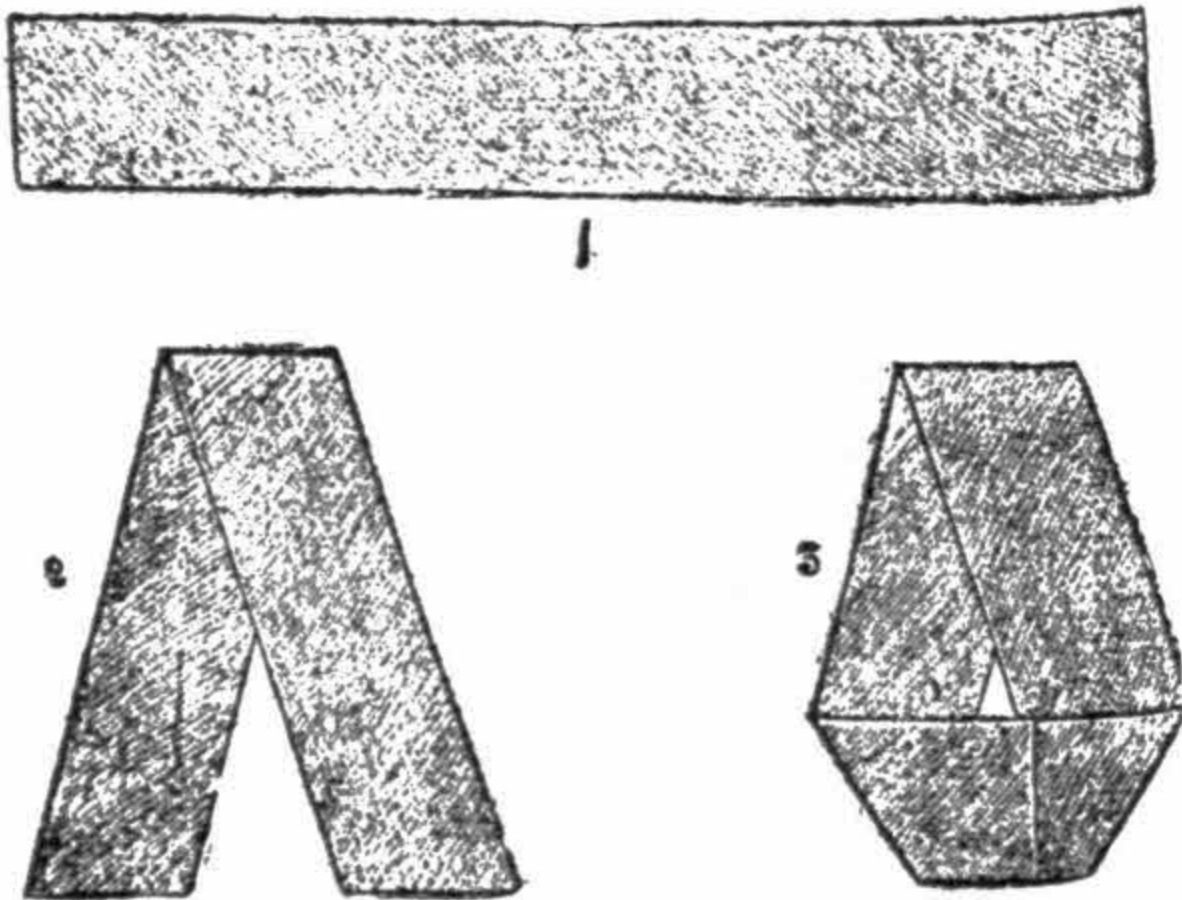


The blades sustain themselves and you may place on them a dish or any other heavy object, without being afraid of a collapse.

The arrangement is sufficiently shown in the design without requiring more detailed explanation.

Geometrical Paper Band.

Take a band of paper, say a postal wrapper; you observe that it has two lines and two surfaces (interior surface and exterior surface.) The problem is to arrange it so that it presents only one line and one surface. It may seem improbable, yet it is possible as you will see. Cut the band and gum together again the two pieces thus separated, after having turned over one of them as shown in figure as above. Arranged in this manner the paper has but one line and one surface, for it has the aspect of a screw without end.



Photographic Camera.

Here is a simple way to construct a camera for a pocket photographic apparatus.

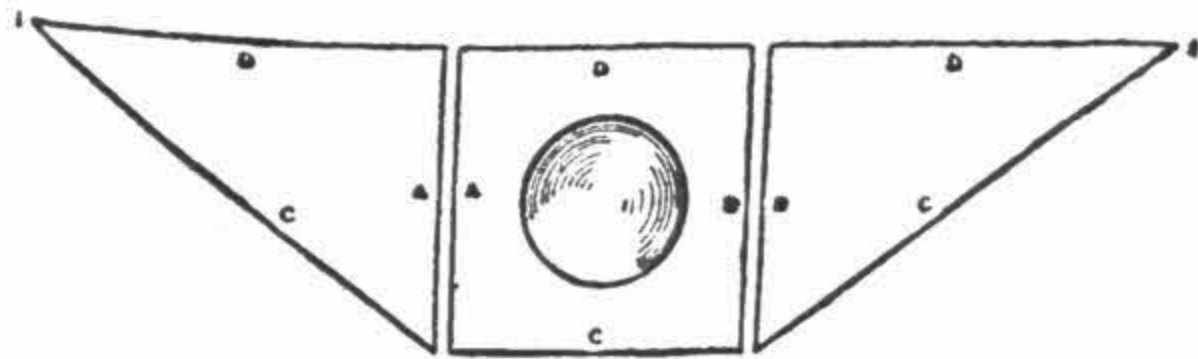


Fig. 1.

Cut out of strong cardboard a piece of about 2 to 2 $\frac{1}{4}$ inches square. In the middle cut out a circle a little smaller than the lens with which you cover it, so that this lens holds on the edge of the hole.

Cut out also two triangles of cardboard, having one side equal to the square, and a length in proportion to the focus of the lens; say for a simple lens of 3 inch focus, and one inch diameter, a length of one and a half inches.

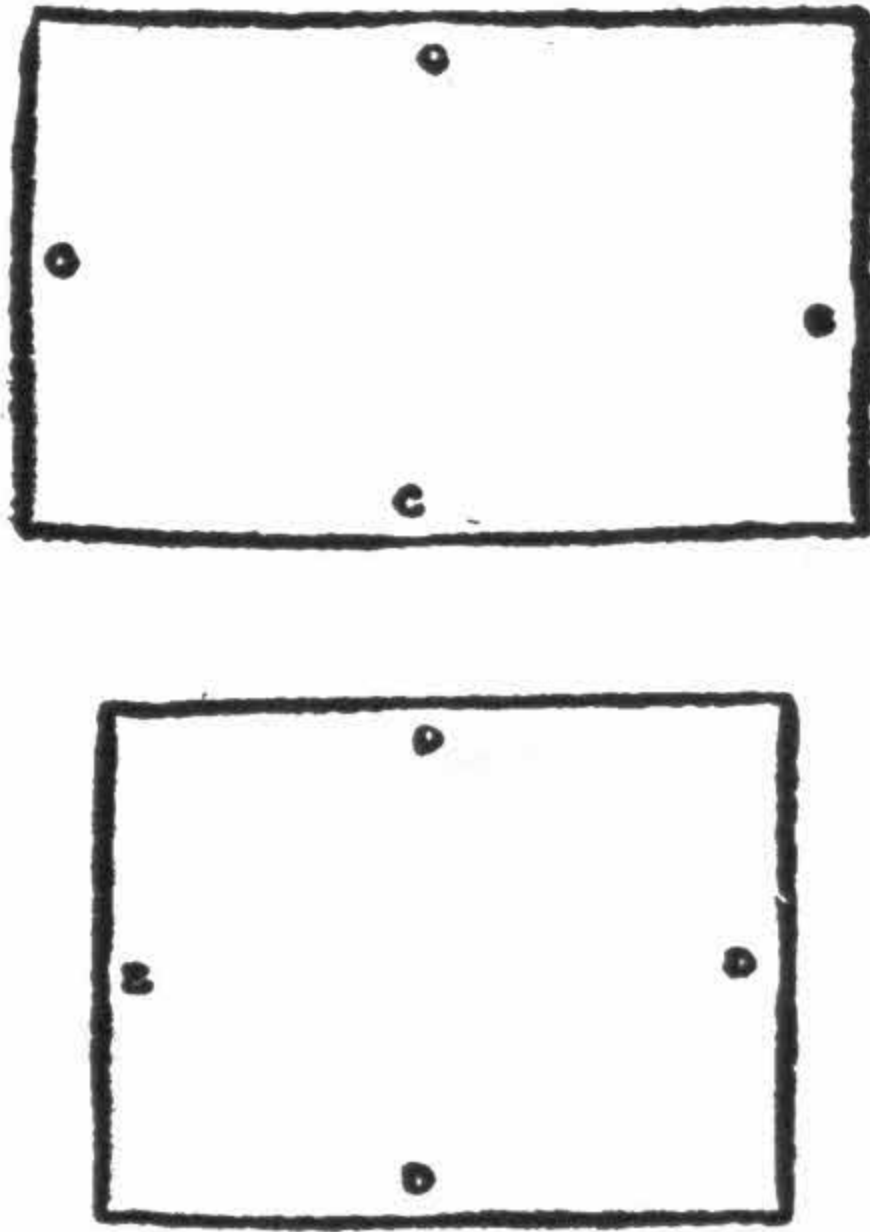


Fig. 2.

Paste the two triangles on the square at A and B, their base C must hold a rectangular mirror of the same dimensions as the side C of the square and the side of the triangles. On side D fix a roughened glass pane, or instead, a thin transparent sheet of paper; tissue paper for example.

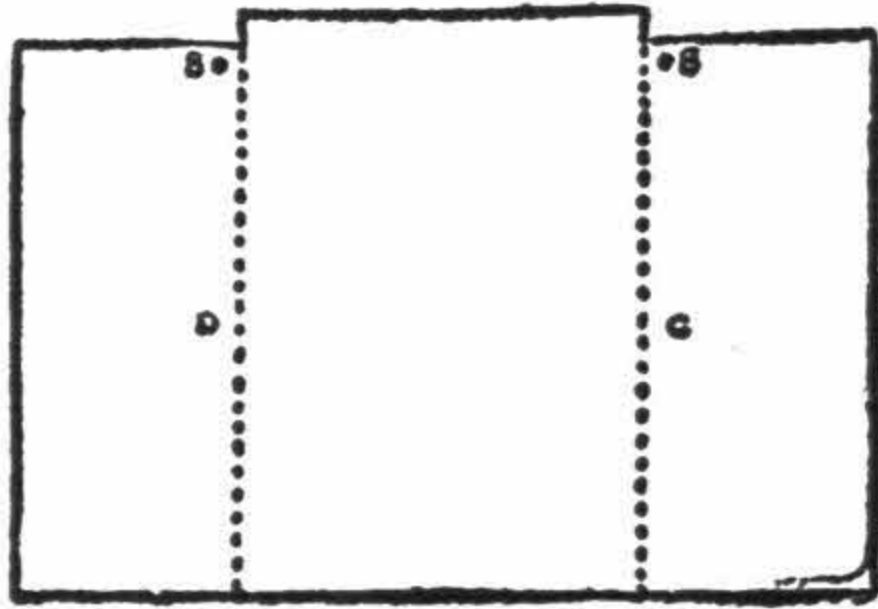


Fig. 3.

Cut a black piece of cardboard as indicated in [Fig. 3 C](#); the dotted lines indicate the sides to be turned down. This shade is fixed on to the camera.

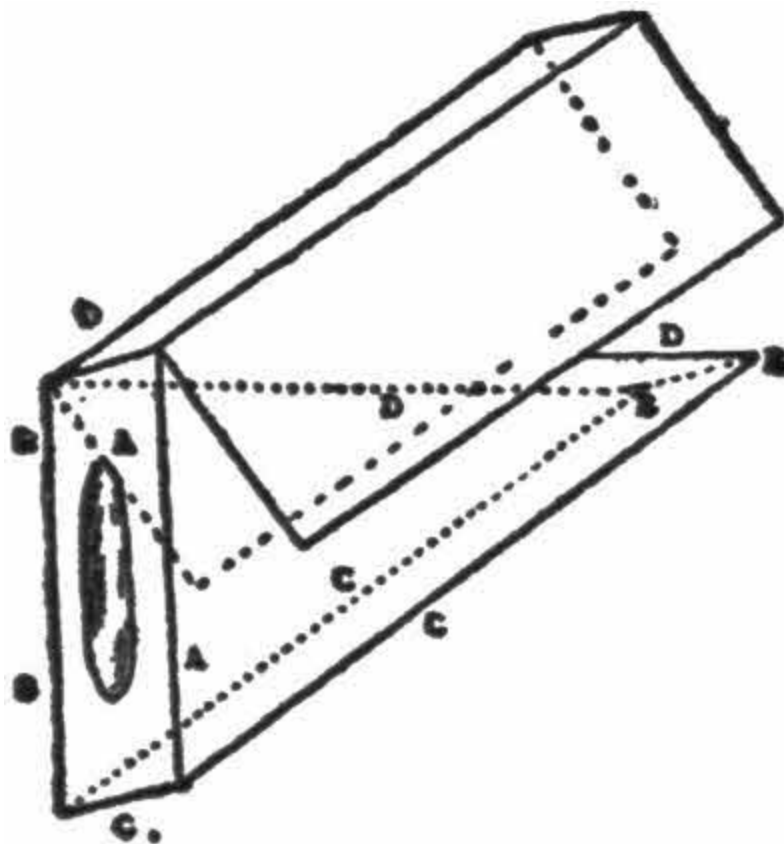


Fig. 4.

Pass through the holes, S S, an iron rod or a long needle, which must pass likewise through the upper angle of the triangles, forming the sides, ([Fig. 1](#)). When your lens has been fixed on the round hole of the square your camera is complete.

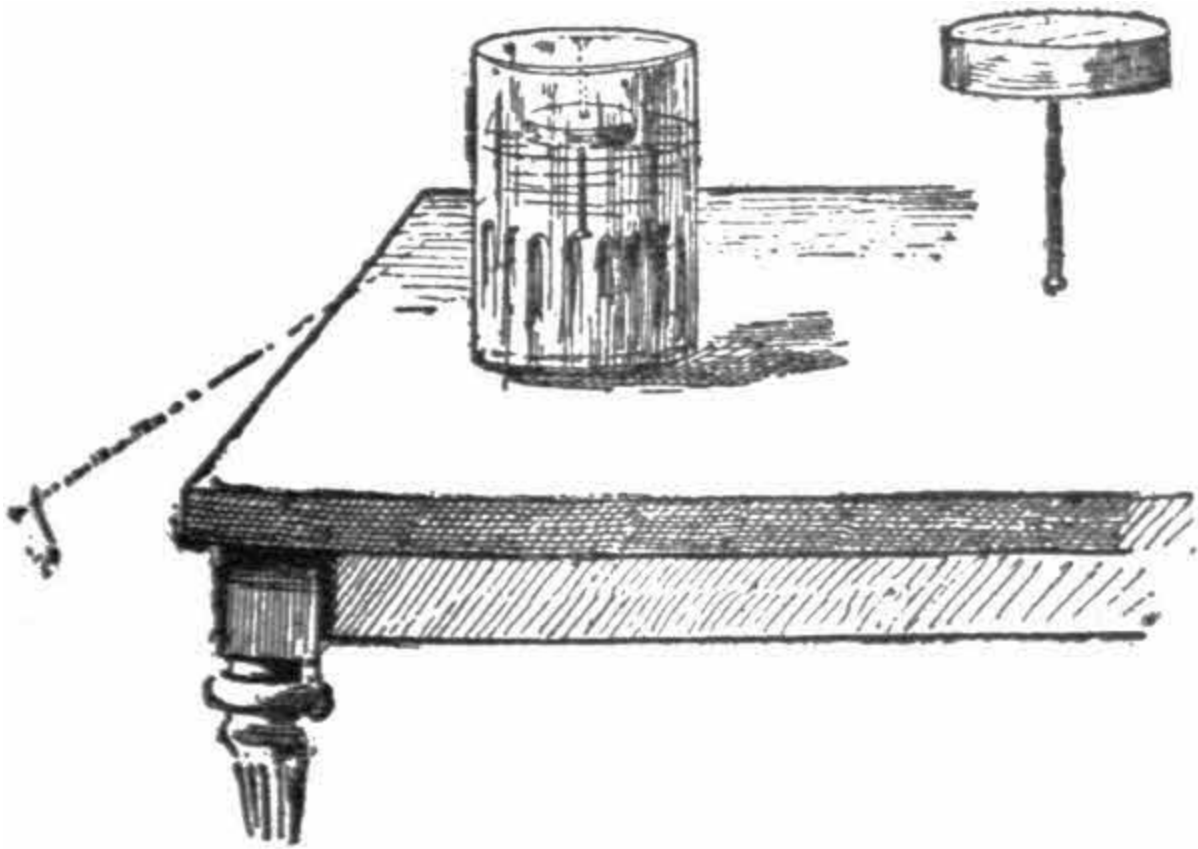
The shade produces complete obscurity so that the operator can see in the middle of the camera the object or person he wishes to photograph.

In order to fix it on the photographic apparatus, one may fasten a wire, in the form of an elongated U, just below the mirror at E.



The Phantom Needle.

You know that when you sit at a window with a looking-glass in your hand, you can catch a beam of sunlight on the glass and throw it into the eyes of a person on the other side of the street.



What have you done in this case? You answer at once that you have bent the sunlight out of its course and turned it in another direction. If the glass were not there it would fall in a straight line on the window seat. This bending out of the straight line is called reflection.

Now for an experiment; cut a small round piece of cork, not quite half an inch thick. Run a needle into its center and place it in a tumbler two-thirds full of water, needle downwards.

Looking down on the cork you cannot see the needle. Now alter your position, and stoop down so that your eye is on level with the table on

which the glass stands. Then you will perceive the needle to be on the top of the cork.

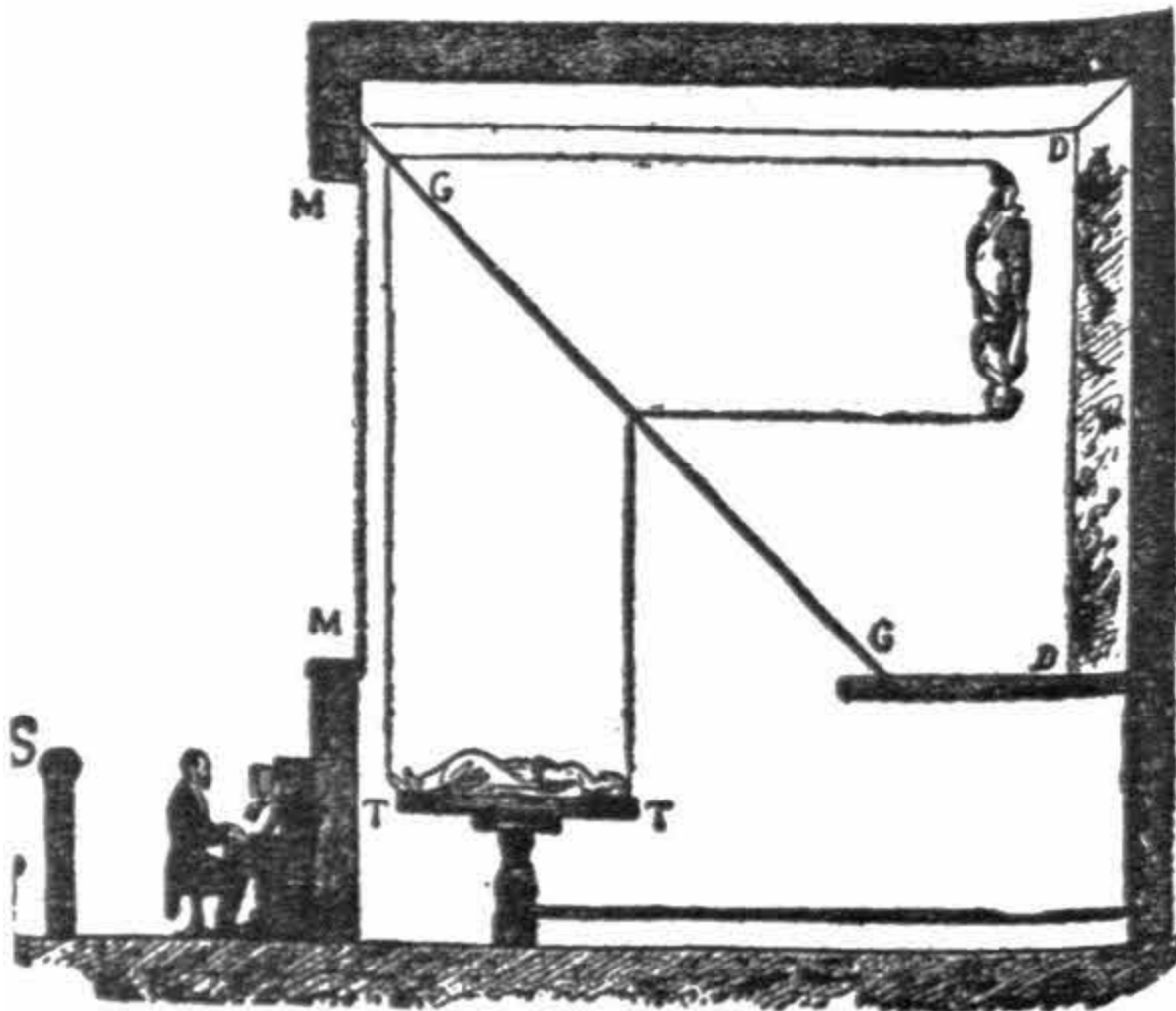
This apparent topsy-turveydom is called total reflection. The needle is reflected on the top of the water, and as the ray from your eye meets the top of the water, you see the needle, as it were, on the top of the cork.



Amphitrite.

At fairs, and in halls of mysteries a variety of optical illusions are presented. Under the name of Amphitrite, the spectacle is sometimes of a woman who seems to rise from the deep, moves about in the empty space, apparently without being sustained by anything or anybody.

She seems completely isolated in mid-air. She turns about, sometimes in a circle, moving now the legs, then the arms. Then after several graceful evolutions in all directions, she stands straight and descends rapidly, seemingly precipitated into a decorated scenery representing the ocean.



The illusion is produced in this way: Behind a well-stretched muslin curtain, M M, is painted D D, with the sky and clouds, below a canvas representing the sea. In front, in the direction of G G, is a mirror, without quicksilver back, inclined at an angle of forty-five degrees.

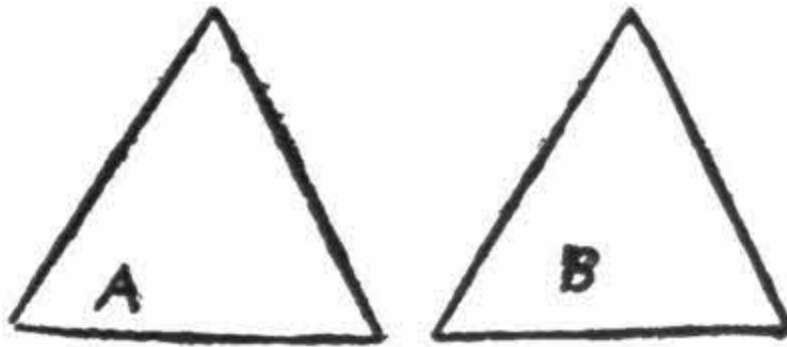
Below the mirror is a round table moving on a pivot, and on this the actress, who takes the part of the Amphitrite, lays down.

In executing various movements, the table in turning, reflects in the glass the image of the person on whom a vivid light is thrown. The spectators placed at S see the image on the canvas at the back, D D. When the time comes for making the lady disappear altogether, the table, which glides on rails, is drawn off the stage, and Amphitrite seems to plunge into the waters. It is by this process that the specters and ghosts at the theaters are produced.

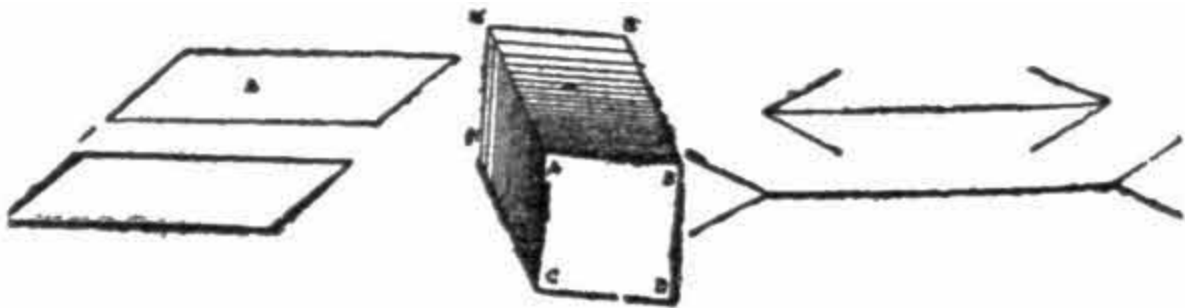
You can perform this illusion, based on the reflection of the light at home, in reducing its construction to the simple proportions of a small theater of marionettes.

Optical Illusions.

Illusions of the eye are numberless, and afford a wide field for experiment. For example, if you ask any one wearing a silk high hat, to what height he thinks his hat would reach if placed on the ground against the wall or door. Nine times out of ten the mark of the height guessed will be half as much again, at least a third over the real height of the hat.



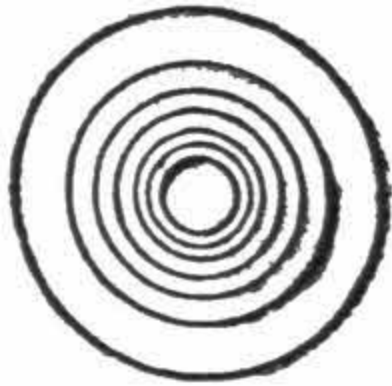
Again, represents two triangles. Ask which is the one whose center is the better indicated. Every one will say, “triangle A.” Well, every one will be wrong, it is B. Take a pair of compasses and you will easily prove it.



The same occurs with the above figure. The two parallelograms, A B, are absolutely equal, and yet A appears to be larger than B. The two lines, A and B are both of equal length; yet B seems a third longer than A.

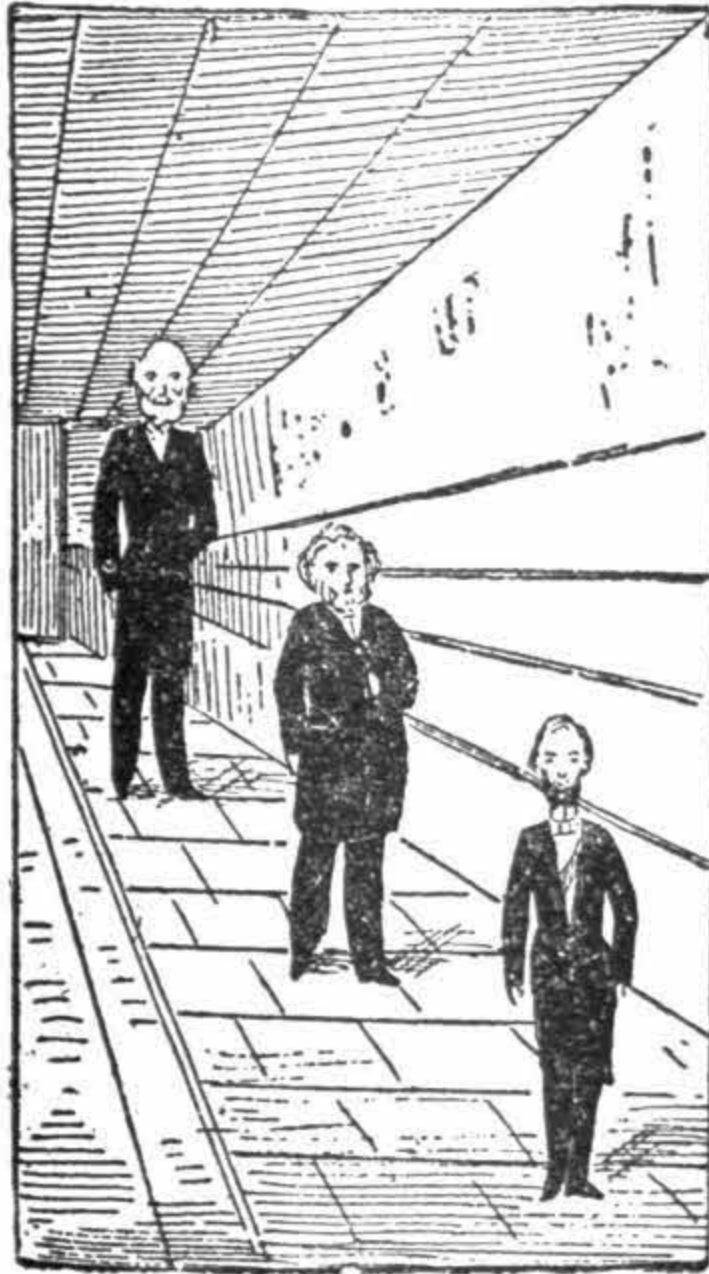
The sides, AB, CD, BD of the middle figure, BE, AM, EM, etc., are equal, yet it seems to the eye that the surface, A B E M, is longer than the square A B C D.

There is another deception the eye is liable to. On a sheet of paper trace several circles, having the same center. Place the sheet on your thumb and turn it horizontally, it will then seem to you as if the rounds turned, though you watch with the utmost attention, the illusion will be complete.



In order to terminate this series, which can be varied infinitely, we will, in our turn, ask you this question: Which is the tallest man of the three personages appearing in the adjoining figure? Is it the first, the last, or the middle one?

Try to find out without any instrument of course, simply by the aid of your eyes which you suppose exact and true. It will appear to you at first sight that the artist has made a mistake, and has made a bad drawing. The last *seems* the tallest, whereas the first seems shortened.

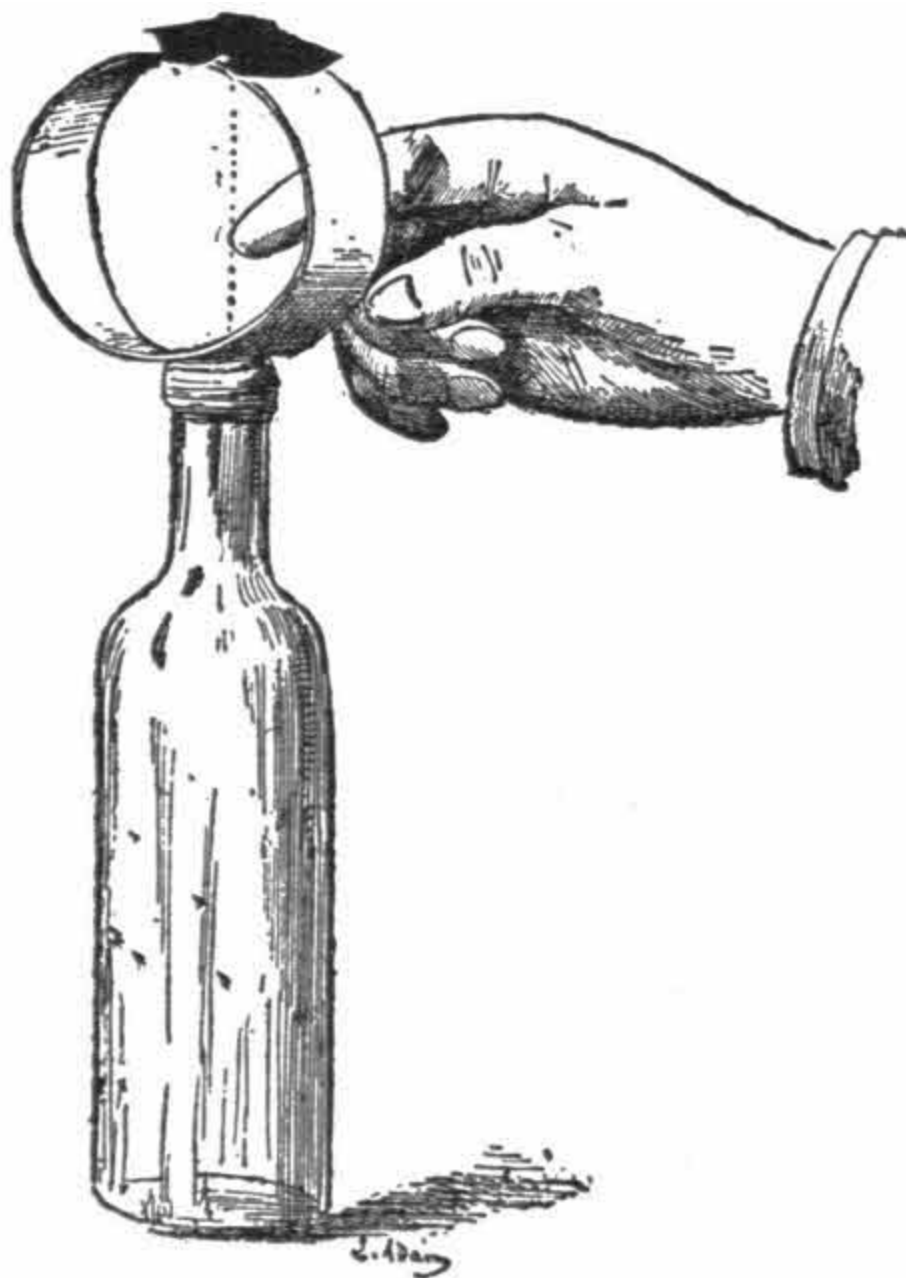


However, measure with a pair of compasses, and the illusion will at once disappear. The draughtsman was not mistaken; the first is the tallest, and the two others go diminishing in height.

This terminates our experiments on optical illusions and you will now enter upon another field of knowledge altogether.

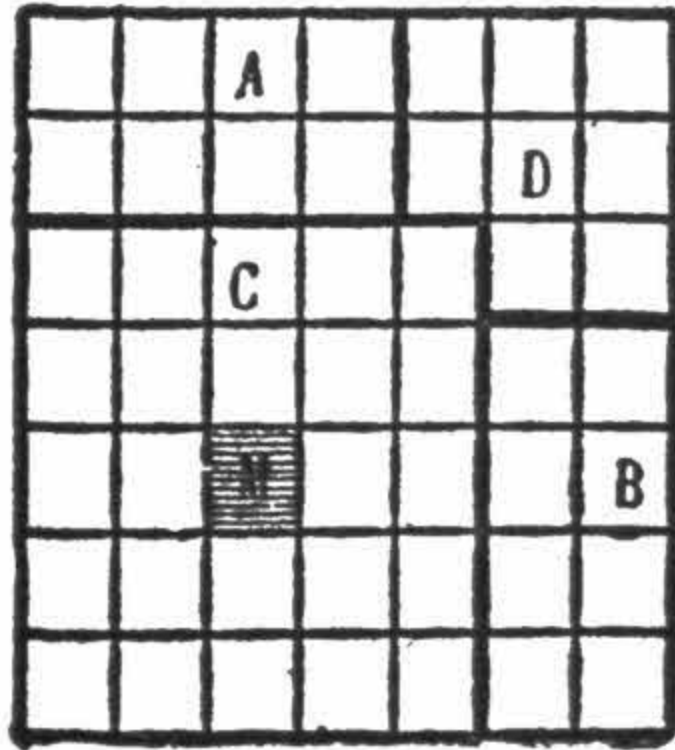
The Insensible Coin.

Cut a piece of cardboard about six inches long, and by sticking the extremities together with a pin, or with gum, form a circle or ring. Balance it carefully on the neck of a wine bottle or decanter, and on the top of the ring place a dime, exactly over the neck of the bottle. Now the trick to be performed is to take off the ring so that, without touching it, the coin falls into the bottle. On the inner side of the ring give a sharp knock with the finger, or, better still, with the thumb and forefinger, as in shooting a marble, as shown in figure. The ring will come off, and the coin which on account of its inertia, does not participate in the movement, will infallibly fall into the bottle. It is absolutely necessary to strike the interior of the circle, because in striking it from the outside one would not get any result at all, on account of the elasticity of the cardboard.



The Asses' Bridge.

Every schoolboy knows which is the famous geometrical theorem, commonly called the Asses' Bridge, and which is propounded as follows:



The square constructed on the hypotenuse of a right angled triangle is equivalent to the sum of the squares constructed on the two other sides.

If we had only to propound this terrible theorem, it would be an easy matter, but the question is to prove it by A and B, and by means of the triangles, similar angles, equivalents, etc. Well, instead of all this, we give here a very simple way to prove the truth; if not quite pedagogic, it is none the less real.

Trace on a piece of cardboard or thick paper a square, and divide into 49 parts. This done, cut it out in following the big lines. Take out on the center one division, which add to the small square, and then construct the [figure 2](#).

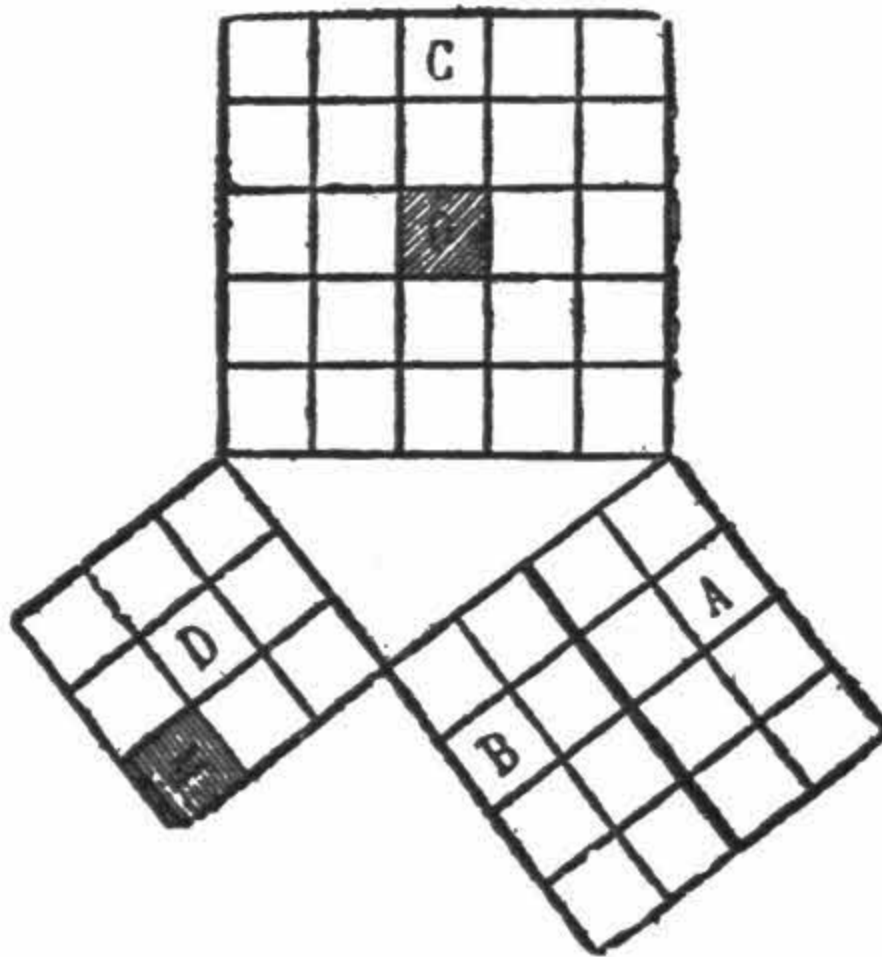


Fig. 2.

The right-angled triangle A C D will be found by the sides of the three squares, and the sum of the two small squares constructed on the two sides of the triangle will be equivalent to the great square constructed on the hypotenuse. Effectively:

Square No. 1 has 9 divisions.

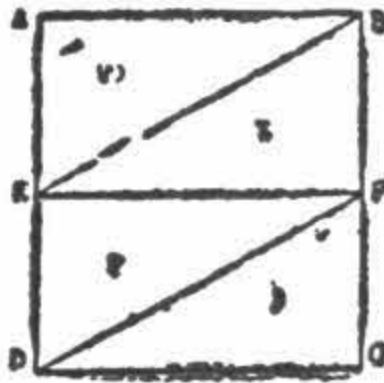
Square No. 2 has 16 divisions.

Together 25

And the square No. 3 has also 25 divisions. Therefore the theorem is proved.

Another Way to Prove the Preceding Theorem.

In a square $A B D C$, trace four similar and equal triangles; cut them out and dispose them as shown in [Fig. 1](#). You will have in the middle an empty space forming a great square, which just has one of the sides of the hypotenuse of the right-angled triangle $A E B$.



Trace the outlines of this square and remount the triangles one against the other, $H C E$, against $A E B$, and $C D G$, against $B F G$, you will get the Fig. below.

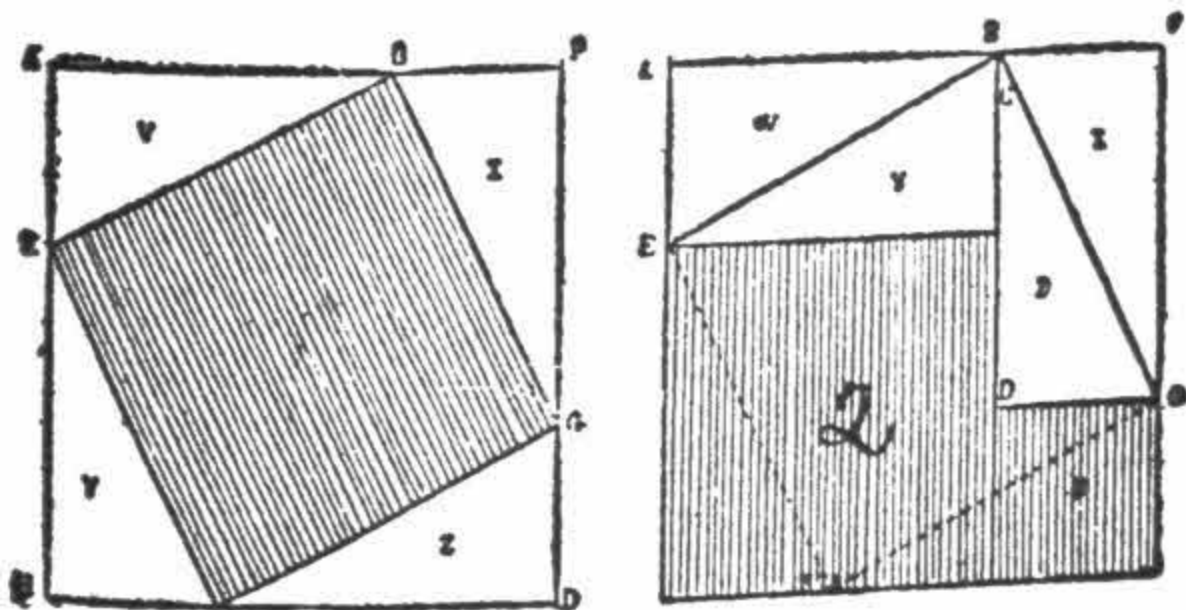


Fig. 1.

The successively covered and uncovered parts of the two squares have not changed in extent. But this time the uncovered part is formed of the two squares 2 and 3 which correspond to those constructed on the two other sides of the triangle, A E B.

This very simple demonstration has the advantage of being applicable to any rectangle.

Indented Angles.

Given two sheets of paper of the same size and form of a rectangle, fold them both in four equal parts, one lengthwise and the other sideways, as shown below.

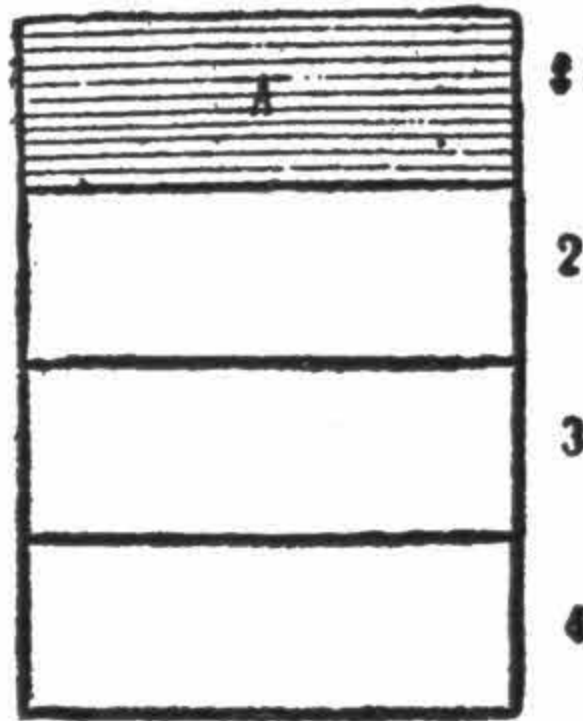


Fig. 1.

When so folded take a fourth part off each. Part A in the figures. The question is now to cover quite exactly one of the remaining surfaces with the other, in cutting the latter in two perfectly equal parts.



Fig. 2.

To resolve this question take the surface (parts A having been detached), with which it is intended to cover the other, fold it again into equal parts, but this time in the opposite way to the one in which it was folded first, as indicated in Fig. 4: cut it out then in following the dotted line, F L, formed by the marks of the fold; this done one will obtain two parts absolutely equal, F L.

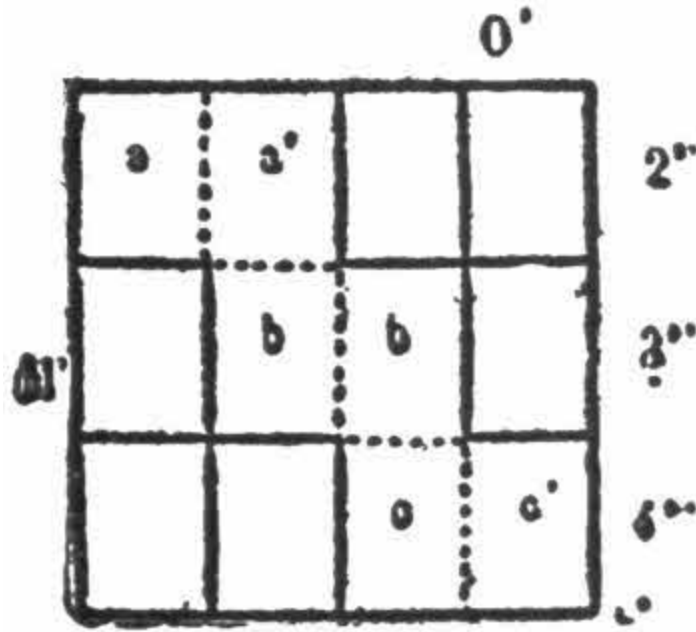


Fig. 3.

In order to cover the other surface, Fig. 3, all that is now necessary is to lower the angles, viz.: angle A', must be in front of angle A, angle B' in front of angle B, and angle C' in front of angle C. When the angles are lowered in this way, the two surfaces will be quite similar, and can be covered one by the other.

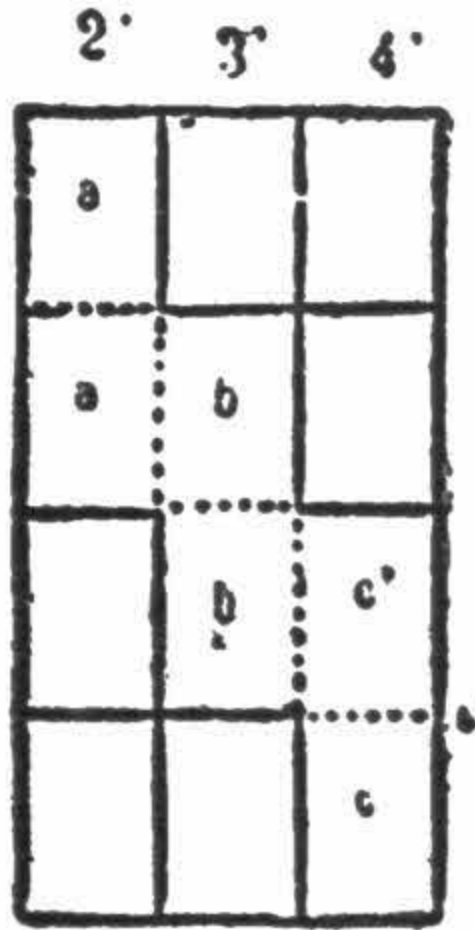


Fig. 4.

This experiment can be made either with one or the other sheet, in lowering or raising the angles.

In the example shown here it is the fourth Figure which is destined to cover the other one.

When the operation is terminated as indicated above, part M, of Fig. 4, will be at M' of Fig. 3, and part O at O' of the covert figure.

A Cheap Shooting Gallery.

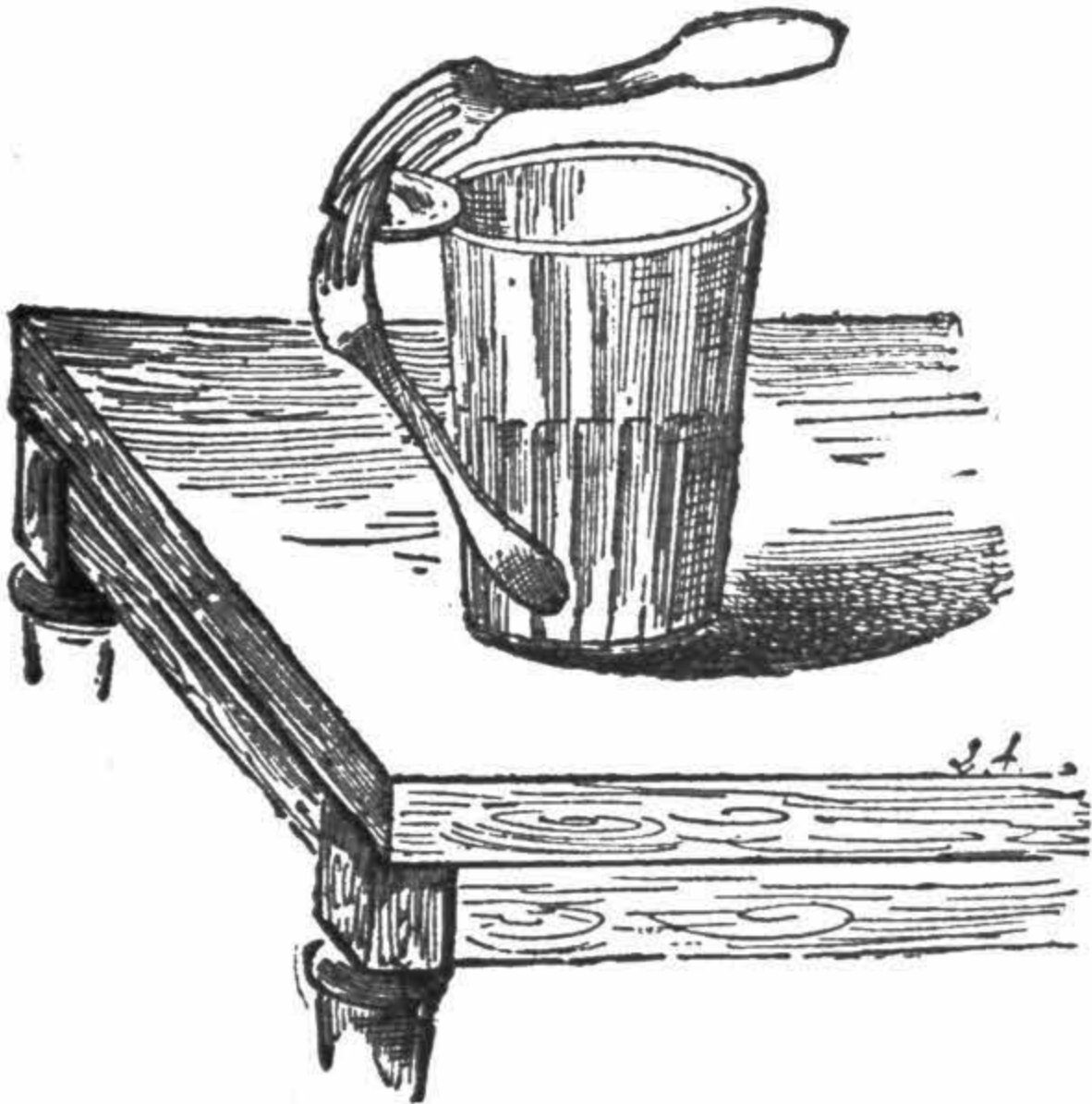
With a whalebone stay busk, make a bow and draw a target on a card. For the arrows, divide lengthwise a steel nib, choosing long shaped ones in the form of a lance and fix each part at the end of a match. You now have complete a saloon shooting gallery, inoffensive and sufficiently recreative at least for your smaller friends.



The Coin in Equilibrium.

Here is a curious demonstration of the balancing of bodies having their center of gravity displaced by a counterpoise.

We propose to keep a coin horizontally in equilibrium on the rim of a tumbler, and it must rest on the glass only by its extreme edge, as shown by the figure which gives the complete demonstration.

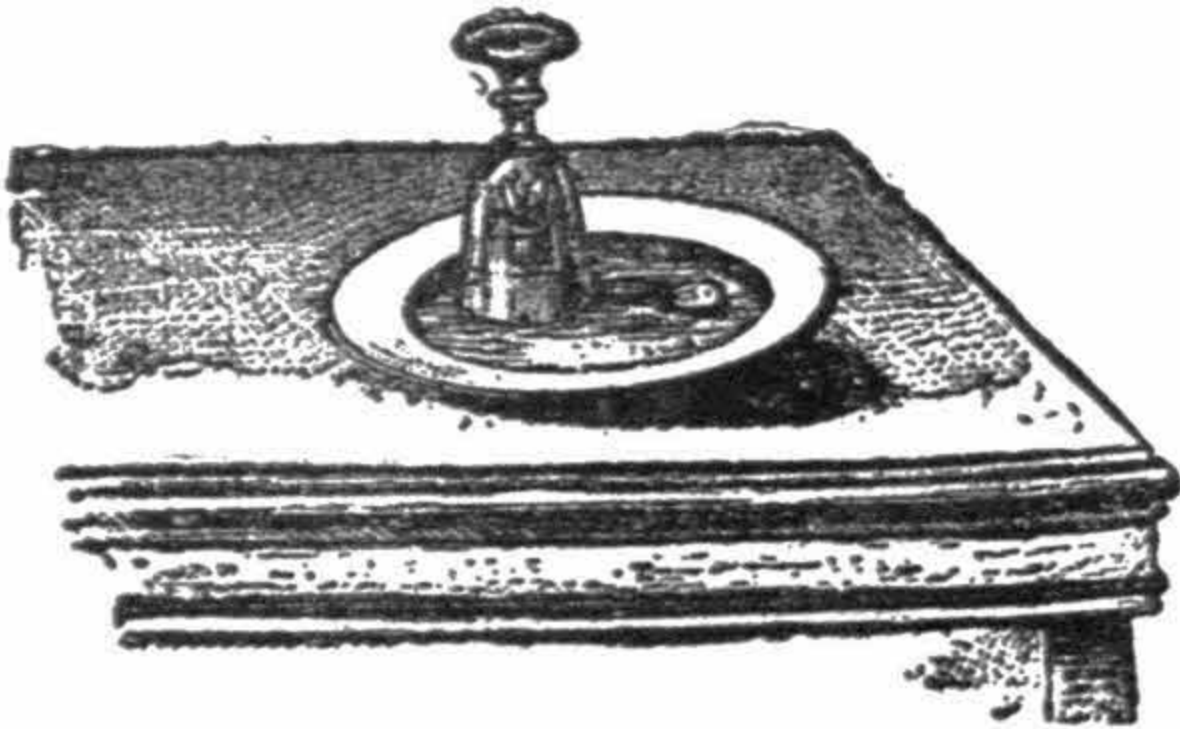


Take a silver dollar and place it between the prongs of two forks covering each other, then place the edge of the coin upon the glass and draw the handles of the forks together, or distend them till the whole are balanced. The center of gravity will then be at the point of contact, and you may give a slight swing without the risk of breaking the equilibrium thus obtained.

The Submerged Coin.

In order to make the previous experiment more significant, you may present it also in the following manner: In a soup plate place a coin; beside the latter an inverted glass, then pour water into the plate just to cover the coin. You then inform the spectators that you will withdraw the coin from the plate without wetting your fingers. You will meet with a great deal of disbelief from many of your friends looking on. Leave them in doubt as to the success of your operation.

Cut a round piece off a cork, on the top of which place some pieces of paper and matches, push the whole underneath the glass, light the matches and wait. As soon as the combustion is over, you will see the water leave the plate and enter the glass, wherein it rises, leaving the piece absolutely dry at the bottom of the plate. You can then execute what you offered at first—take out the coin without wetting your fingers.



As a variation of the preceding experiment, obtain a flat-bottomed tumbler or glass goblet (but the bottom must be flat), a pocket handkerchief and a

coin. These are to be seen by everybody present.

Procure a watch glass, or a round piece of glass like an eye-glass. This is not to be shown.

Now show to the bystanders that you place the coin (say a fifty-cent piece, for example) in the middle of the handkerchief, and, throwing back two sides of the latter, point out again that the coin is still in its place.

To show that there is no deception ask someone to hold the coin in the handkerchief.

Then place underneath it a glass containing a little water and call out, "Hey, Presto! Fly!"

The person lets go of the coin and the noise of its falling to the bottom of the glass is plainly heard.

You take up the handkerchief, and every one is astonished at the disappearance of the coin, which you can produce from another person's hat.

Really the trick is very simple. For the coin supposed to be held in the handkerchief you must dexterously substitute the watch glass or eye glass. The person holding it, of course, declares he has the coin fast.

When he drops the eye glass it makes the same noise against the tumbler as would the coin, though, of course, it cannot be seen in the water.

It is a capital trick if smoothly performed.

The Smoke Rings.

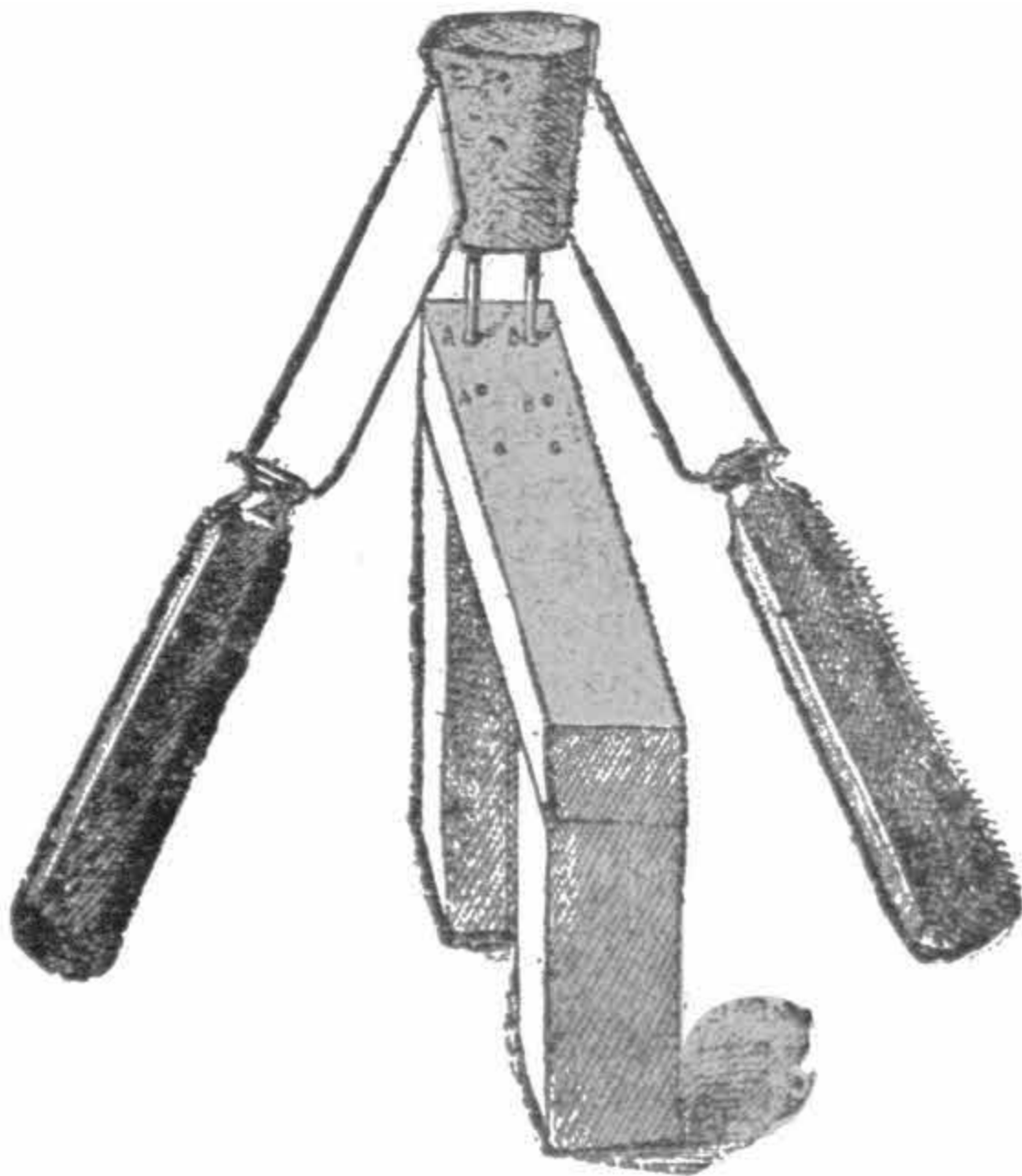
When the air of a room is very calm, have you ever noticed that tobacco smoke rises slowly and in a nearly vertical direction? Have you never watched with interest the grayish or bluish streaks of smoke issuing from the smoke of a cigar or pipe? And in seeing the smoke rise in such a capricious fashion, have you ever ascertained that it is due to the calm of the surrounding air?

One may get some amusement out of the agitation of the atmosphere. For the materials you only require a square or round cardboard box, in the lid of which cut a round hole of about two inches in diameter. In the interior of the box place two sheets of blotting paper, the one impregnated with muriatic acid, the other with ammoniac, in equal quantities. Immediately a whitish smoke will escape through the hole and rise straight to the ceiling. If with both hands you give a series of simultaneous taps on the sides of the box, you will see the smoke issue in well defined rings, which will disperse rapidly in the air, and succeed each other as long as you continue the pressure. These rings are the result of the concussion of the air which you occasion in the box. The experiments can be made also with ordinary tobacco smoke, but it will last longer in the way we have indicated.



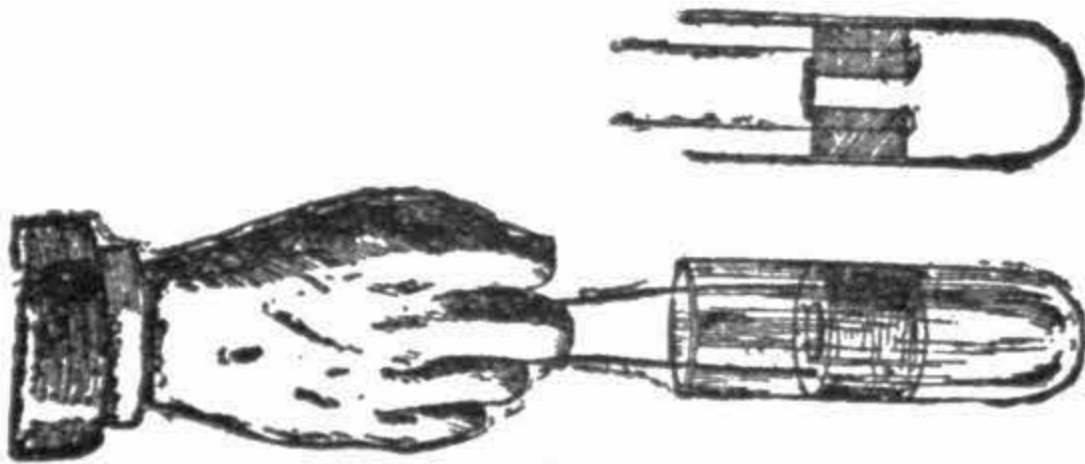
The Walking Cork.

Stick two knives in a cork, on the same level, and opposite each other, so as to form a balance. In the bottom of the cork, at an equal distance, insert two pins, sufficiently deep not to bend under the weight which they will have to carry. Rest the pins on a flat ruler, slightly inclined, and give them a slight balancing movement. The weight of the apparatus will fall on the pin, A, on which the whole turns the knife placed at the side, B, will knock against the support, and will tend to bring the apparatus again into its original position as the oscillating movement continues, pin, B, will, in its turn, support the whole weight, and pin, A, will shift on to the other point, A, indicated in the figure. The walking cork will thus continue its movement till it has gone over the course assigned to it. This recreation is interesting, for it demonstrates once more that all bodies are attracted by the earth, and that, as soon as they are thrown off their balance they obey the force which constantly draws them down.



The Obstinate Cork.

Take a glass or metal tube, closed at one end, and cut a stopper in cork or india-rubber to its size so that it closes the tube hermetically, and take care that it glides in the tube without difficulty, and pierce it with a hole. On the top of this hole adapt a small piece of leather, rather larger than the hole, which you must take care to wet before proceeding with the following experiment. In order to be able to withdraw the cork when in the tube, you take the precaution to fasten in it both ends of a piece of string, as indicated in the figure.



These various objects being prepared, will serve to demonstrate once more the atmospheric pressure.

Lift the valve up, force the cork into the tube towards the middle, and when there, put the valve in its place again, and pull quickly on the string as if trying to pull the cork out. The latter will not come out, for the reason that producing a vacuum before itself, the atmospheric pressure will prevent it from coming out. But if instead you draw it gently towards you, it will offer much less resistance, because the outer air will enter through the smallest interval between the cork and the glass, and partly destroy the exterior pressure.

Petroleum Pulverizer.

With the aid of the compressed air reservoir you are able to conduct various soldering operations, requiring often great heat.

Let us construct the following pulverizer:

Into a bottle of the shape shown in the figure, put some petroleum, and introduce a glass tube that does not quite reach the bottom. Close with wax that no air can enter, and at the upper extremity of the tube let a fitting be embedded, a section of which is shown in the engraving. This fitting has three openings, two horizontals and one corresponding with the tube that is plugged into the petroleum.



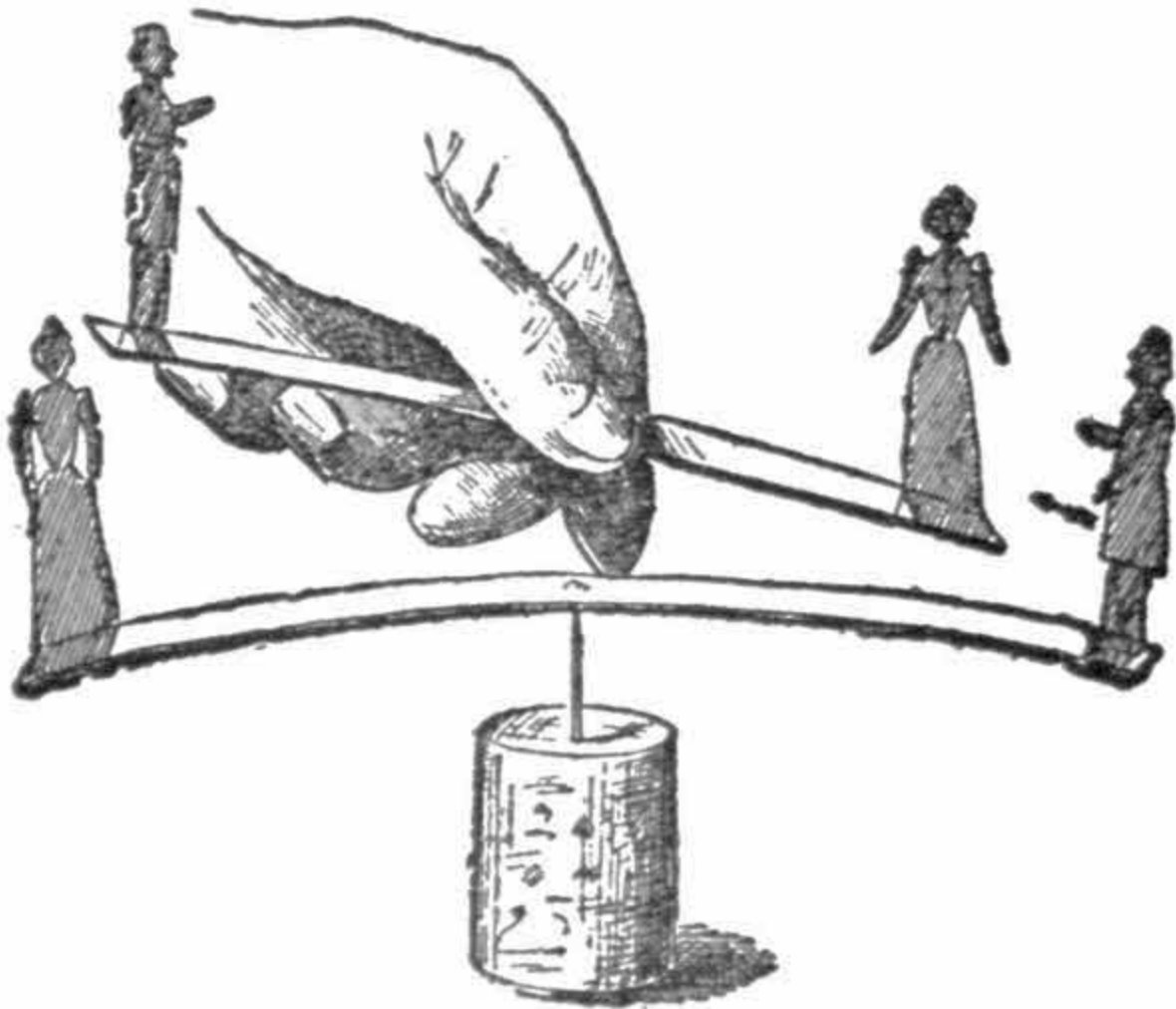
Adjust the india-rubber tube to the reservoir, and when the pressure is exercised on the surface of the liquid it will force the petroleum upwards through the tube, and thence it will be blown in a fine spray, which burns as if coming out of an ordinary burner. The particles of oil are mixed with air, and consequently the atoms of air are heated to a high degree. This jet develops a heat of great intensity.

This pulverizer may serve also to disinfect rooms. You have only to replace the petroleum with an antiseptic liquid.

Electric Attraction and Repulsion.

The poles of the same sign repel, and the contrary poles attract each other, or, in other words, the negative, or the positive electricity attracts the electricity of a contrary sign, whereas the electricities of the same signs repel each other.

In order to demonstrate this principle we will contrive a little plaything which will be as interesting as amusing to see in operation.



For a pivot take a needle stuck in a cork, and, as magnetic needles, two old corset steels will do very well. If these cannot be had a clock spring may do

instead.

Magnetize these two steel rods by rubbing them with a magnet. In the middle of one of the rods punch a small hollow so that it may freely move on the needle in the cork without fear of falling off. You have thus manufactured a rough compass.

Then cut out four dolls in paper, two gentlemen and two ladies, and stick them in the extremities of the two magnetized needles. Remember to put at each end a figure of the opposite sex.

Now, each time you present a man to the other man, which is placed on the magnetized needle, they will repel each other; if to a lady, the dolls are attracted.

The explanation is easy. You will have taken care to put the puppets on contrary poles: a man on the positive pole, a lady on the negative pole. In this way the principle enunciated above is thoroughly proved and easily grasped.

One may easily vary this experiment by replacing the gentlemen and ladies by personages of actual notoriety, or of the company, in placing them in groups which have a mutual dislike to each other, such as the schoolmaster and pupil, etc.

The Bust of the Sage.

Every person wonders how the sensational decapitation scene is produced.

To all appearance a head is thrust through the neck opening of a guillotine, the knife descends and the head is cut off. However, in order that none of the fair sex may be alarmed, it may be varied as follows:

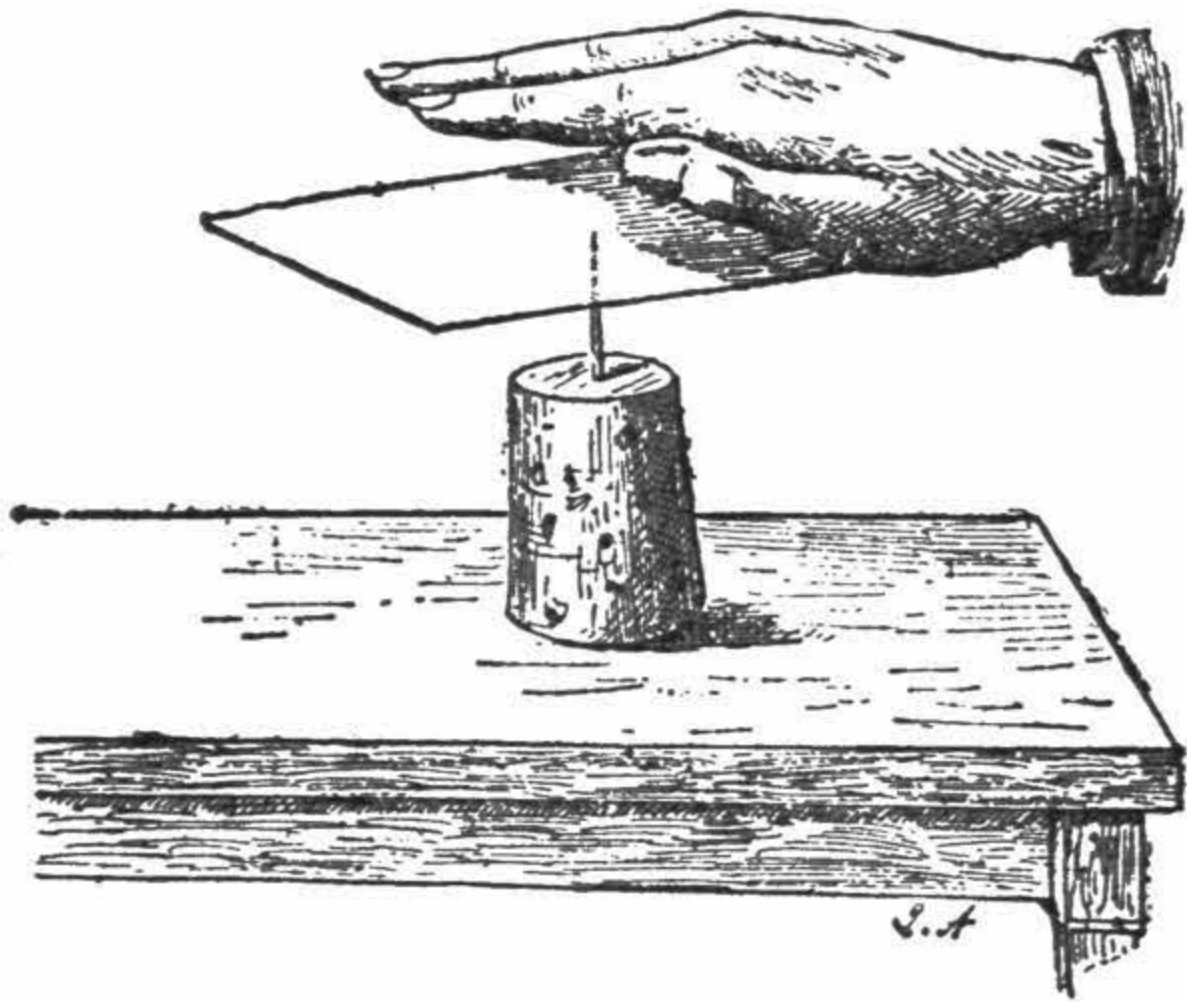
In a cabinet a mirror is set across, sloping from the top at the back to the front. It reflects the ceiling, which is covered with the same material as that which is seen of the floor in front of the mirror. In the center of the glass is cut a hole which admits of the passing through of a man's head. He sits or stands in under the glass at his ease. The front edge of the mirror is concealed by a few astronomical and geographical instruments, old folio books, skulls, etc.

The head and shoulders are made up and draped like one of the seven wise men of Greece, and he answers questions in a grave, portentous voice.

The Witchery of the Hand.

In order to ascertain the existence of animal magnetism the following apparatus, very simple, and not at all difficult to construct, answers perfectly.

Stick a pin in a cork, point upwards. On that pivot place horizontally a sheet of paper so that it remains in perfect equilibrium.

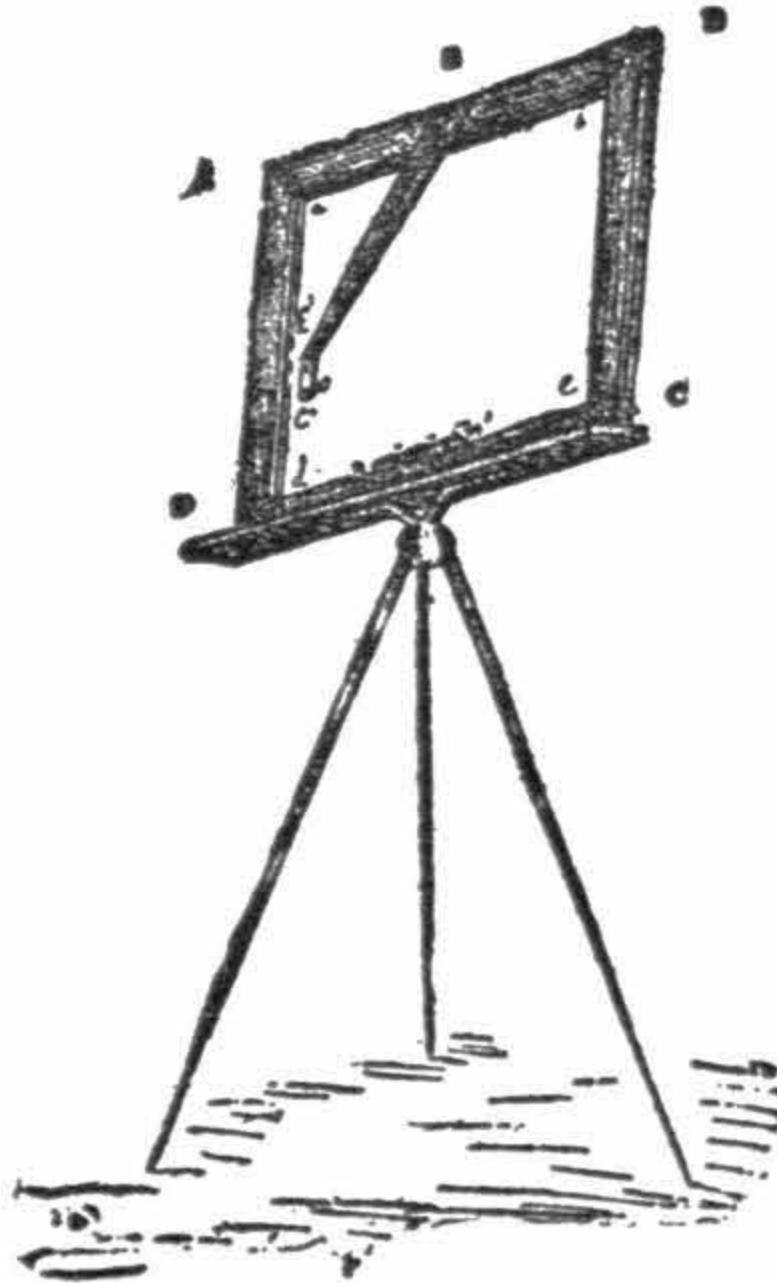


If you now put your hand over the sheet of paper, a rotary movement will manifest itself, the sheet swerving from right to left. This movement is caused by the influence of the hand's magnetism.

The Perspectograph.

This simple instrument, invented by Mr. Jarlot, renders the tracing of a sketch extremely easy, besides avoiding absolutely faults of perspective, which is, without doubt, the principal advantage of this instrument. Thanks to it, one obtains an easy reproduction on one plane of objects placed on different planes.

Here is a description of this very simple instrument. A wooden frame A B C D, with a slot in the side, A B, in which a pane of glass can slide so as to cover the whole space of the frame, *a, b, c, d*, is fixed on a stand.



The frame is maintained in a perfectly horizontal position by means of a water level $n n'$, placed on the lower side of the frame. At E is a small rule moving on a hinge at E, allowing the angle to be varied at the plane A B C D, by resting it on two supports E E'.

The supports themselves move round on an axle fixed on the rule. At the extremity E' of this rule is fixed a copper blade curved in E', C', and pierced by a small hole of about an eighth of an inch in circumference, the

edges are made thinner as represented in the diagram placed above; the widened part is turned toward the frame.

So much for the body of the instrument, now for the accessories. In the slot left in side, A B C D, lower a glass pane covering the space, *a, b, c, d*, which is not, however, a necessary condition, it depends on the size of the design you desire to take.

This pane requires a little preparation. It is done in this way: One chooses a pane of the desired glass, as free as possible from defects. Cover one side only with turpentine, and which you know is a natural varnish.

See that this coating is as thin as possible, and to ensure this, go over the surface with a very soft brush steeped in the liquid. When you see that the latter does not run any more, leave off brushing. Let it dry for two days if necessary; take care, meanwhile, to protect the varnished side from dust.

Now it remains only to show the use of the instrument. Put yourself in front of the object you wish to represent. Put the frame in a perfectly horizontal position, slide the pane in it, and dispose the rule, E E' ([Fig. 2](#)), in such a manner that, when looking through the little hole, O, you are able to see the object you want to draw.

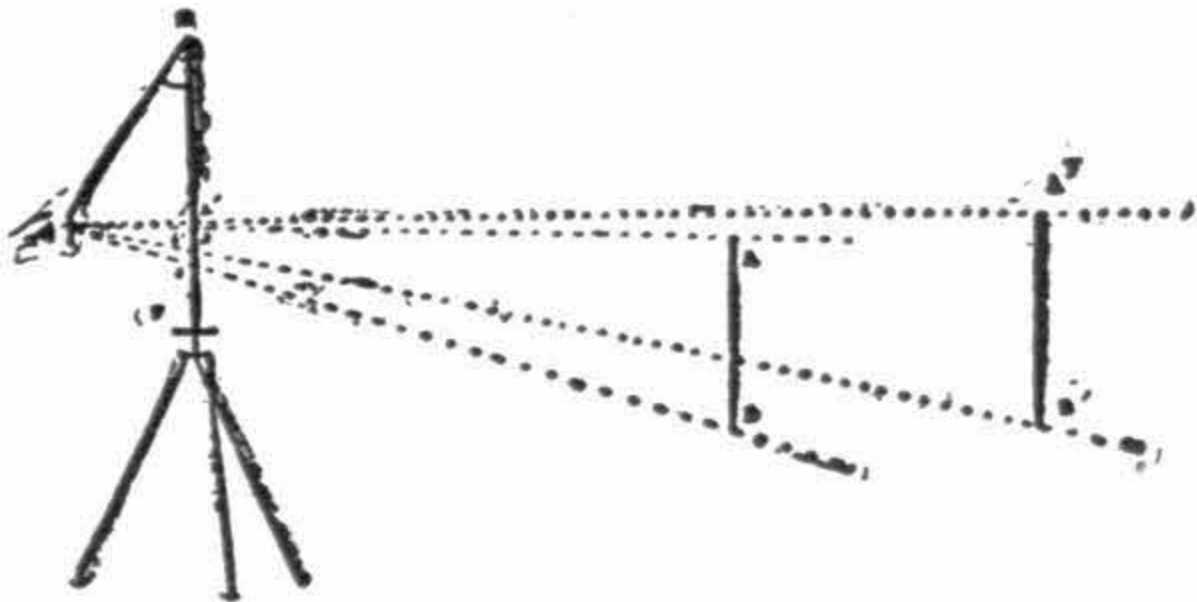


Fig. 2.

Then, with a blue or other colored pencil, trace the outlines of the object on the glass coated with turpentine, the use of the latter being to allow the pencil marks to fix itself on the surface. One sees that the outlines thus obtained will be those of the real object as clearly as possible because they are traced as seen, so to speak.

But the principal object of the instrument is not so much exactness of outline as to get the exact proportion existing between the different sizes of the objects placed in different planes. We will try to show this last result by means of [another figure](#).

Suppose AB to be an object situated at a certain distance from the eye posted at o , the rays from the eye, OA OB , meets the instrument at a and b , and the image of this object is given by the line, ab .

Now, suppose $A'B'$ to be another object situated beyond AB ; the eye has not changed position, it cannot do so, with reference to the glass, on account of the small rule which is fixed; the image of the object $A'B'$, will be $a'b'$; thus, one has the true dimensions of $A'B'$, in respect to AB .

It is precisely this proportion which must exist between the sizes of the objects placed in different planes, which constitute perspective. The instrument, therefore, well deserves its name of Perspectograph.

It will be observed that this apparatus obviates two difficulties: 1, that of the exactness of the sketch, in copying nature as it is presented to the eye; 2, that of perspective. Having the sketch on glass it is easy to transfer it on paper. Lift up the rule, EE' , so as not to be in the way, place oiled or transparent paper on the glass, and counter-draw the sketch on it.

You can then stick this paper on a cardboard, and, if the operator is a designer, he may reproduce in crayon a very fine drawing. For the shading he must use his own talent, the aim of the instrument not being to give a finished drawing, but only a sketch, vigorously exact, and in unexceptionable perspective.

This instrument is often very convenient. When wishing to have a true sketch, you trace it on the glass; you then transfer it to an oil-paper, and again on drawing paper if the former is not to be used. Besides, if one has an exact sketch on whatever paper, you may reproduce it in freehand, if you are blessed with any idea of drawing.

If the varnished plane has to serve again, wash it with warm water, and let it get dry; then the varnishing can be done over again.



Camphor in Water.

If you put very small pieces of camphor on the water, you will see them turn round each other with great velocity. These movements are due to the diminution of the superficial tension of the liquid in the vicinity of the pieces of camphor.

In order to stop them throw a drop of oil in the water, and you will produce a perfect calm. One may utilize the camphor for an amusing recreation.

Construct a small paper or cardboard boat and fasten underneath on the hind part or stern a piece of camphor. Your boat will maneuver on the water. Persons not initiated will be much puzzled, and be long to find out by what contrivance this small craft is propelled.

A Simple Multiplier.

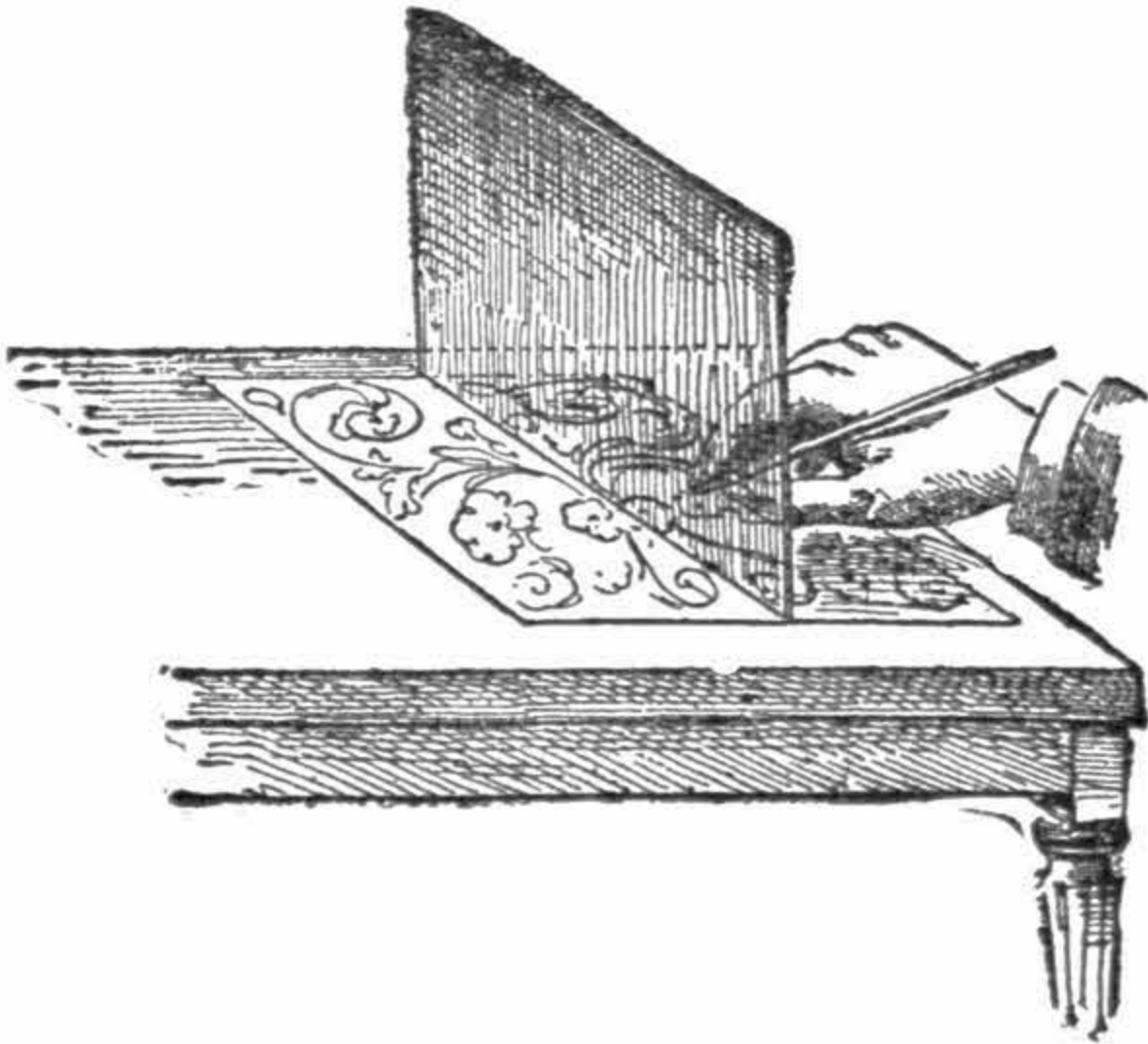
Write on a card or strong paper the letters, figures, etc., which you want to reproduce. Then all along the lines or tracing, with a needle, prick holes in close proximity at equal distances.

Place the sheet so prepared on a pad made of several sheets of blotting paper, smeared with blue analine ink, or a mixture of lamp-black and oil.

Fix the corners with tacks or drawing pins, and draw your copies by simply placing the blank sheets over the pricked one, and press them down. The words, figures or designs will be reproduced in dotted lines if the holes have been well pricked. In this way a good number of copies can be drawn.

The Drawing Room Mirror.

On one of the faces of a pane of glass smear some lamp-black mixed with oil. If you place this glass, thus prepared, vertically on an engraving representing flowers, fruits, birds, etc., you will obtain an infinity of forms, some of which will be very striking.



If you want to reproduce these, to fix their outlines, you have only to interpose a transparent paper, to draw along the glass pane a line in pencil and to trace over the part of the picture which terminates at the foot of the pane.

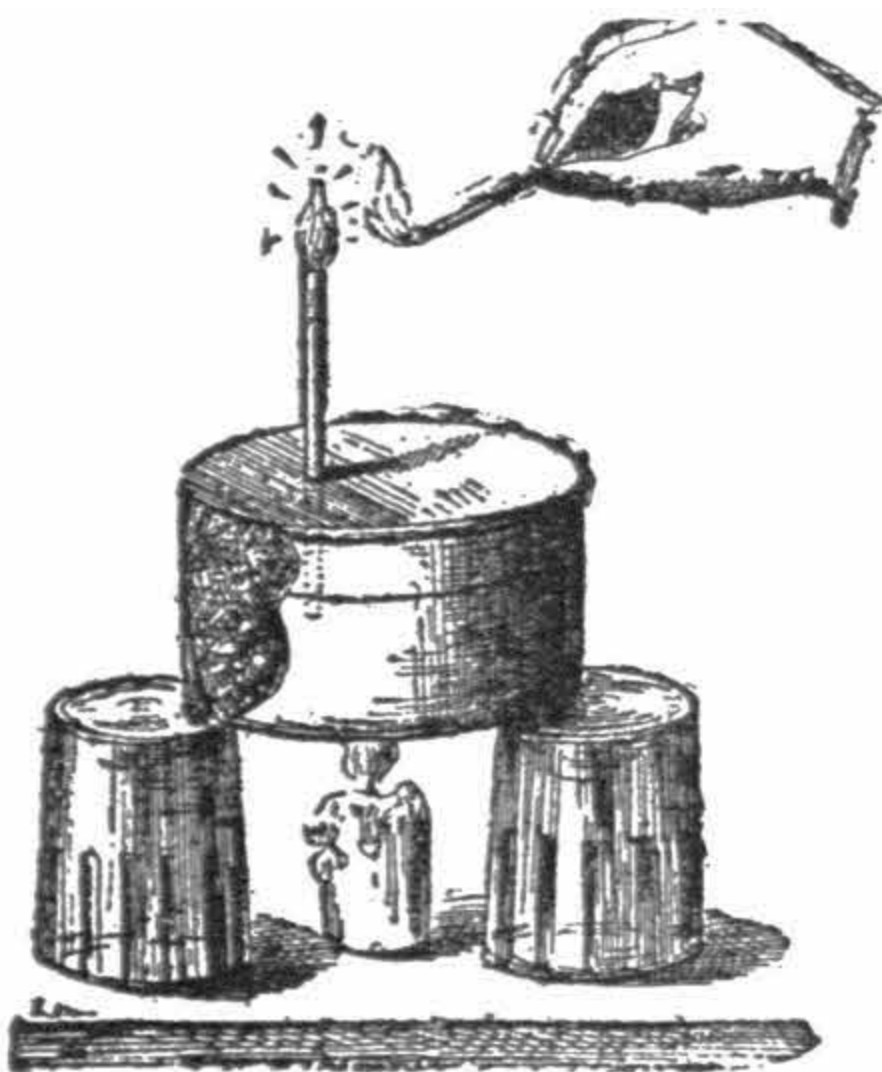
Fold the transparent paper along this pencil line, and to get the whole reproduced you have only to copy over the designs just traced. The glass, which does the duty of a mirror, doubles the forms in a symmetrical way, and as it is moved new forms come to view.



Elementary Gas-Burner.

Fill an old round tin box, at least two inches high, with sawdust and pieces of blotting paper. Close it as well as possible, and introduce a small metal, or glass, tube in the lid to a depth of about one-third of the box.

Make the joint tight with putty. Put this box on any two supports and place the flame of a lamp or candle underneath it.



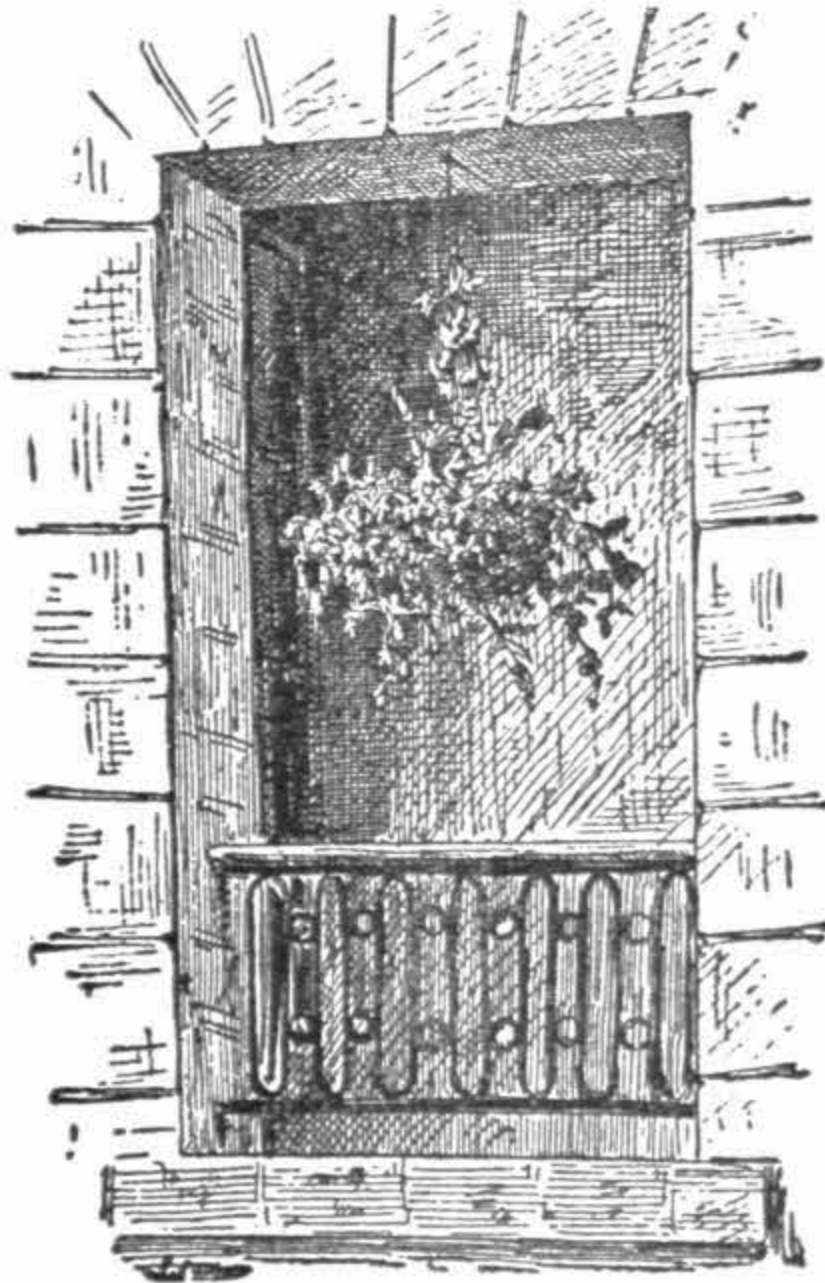
Soon the overheated sawdust and blotting paper will evolve vapors of alcohol and combustible gases. Approach a lighted match to the upper end

of the tube and you will see the gas ignite and continue to burn.



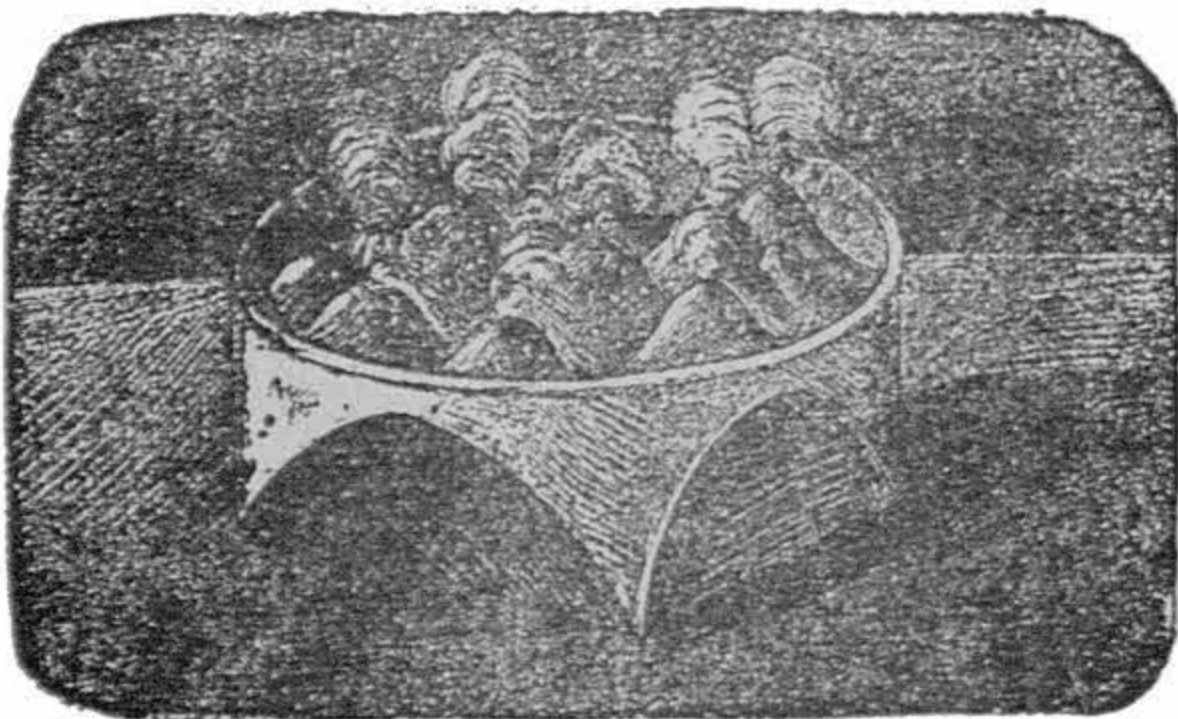
Rapid Vegetation.

A cheap sponge can be converted into a hanging bunch of greenery for room decoration. Plunge it into hot water, press it dry, then put in its holes or pores seeds of millet, red clover, barley, linseed, grasses, etc., in fact any species of plants which germinate easily and produce, as far as possible, leaves or blades of different shades. Place the sponge thus prepared on a vase or in a saucer, or better still, suspend it in the recess of a window, where it may get as much sunshine as possible. Every morning, for a week, sprinkle its surface lightly with water. Soon the seeds will germinate and grow. In a short time they will form a ball of greenery, making a charming decoration for a room.



Miniature Volcanoes.

In a rather large porcelain or glazed earthenware basin place a small quantity of nitrate of lead. This may be obtained for a few cents from any painters' supplies store. Then upon it throw some flakes of sal ammoniac. Immediately a number of conical elevations will be formed, which give off vapors and burst with a popping noise.



Altogether the experiment represents very exactly, volcanoes in a state of eruption. When the eruption ceases, the rough, broken state of the remaining mats, represents as nearly as a thing on this earth of ours can, the appearance of the moon through a powerful telescope.

Indeed the moon at one time of its history was undoubtedly in the soft lava-like state that you will observe in the basin, during the first stage of this elegant and instructive experiment.



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Illustrations have been moved to paragraph breaks near where they are mentioned.

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Variations in spelling and hyphenation were retained as they appear in the original publication, except that obvious typos have been corrected.

Changes have been made as follows:

[p. 3](#): Caption added to figure (Fig. 1.)

[p. 40](#): Caption added to figure (Fig. 2.)

[p. 41](#): Caption added to figure (Fig. 1.)

[p. 56](#): Caption added to figure (Fig. 2.)

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